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# Double Vision: Donor Information, Farmer Voice, and Irrigation Policy Reform in Uzbekistan [*†]

Muzna Alvi
IFPRI–New Delhi
Anton Liutin and Paul Castañeda Dower
University of Wisconsin-Madison
Claudia Ringler
IFPRI

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**Abstract**

This paper studies the channels of feedback on policy effectiveness in a formerly authoritarian regime. We study Uzbekistan's flagship subsidy program for water-saving irrigation technologies. Read from formal sources, the program looks like a success. We then compare five information channels, a pseudonymous farmer Telegram forum with approximately 330,000 messages, farmer interviews, independent and state media, and donor documents, as well as the legislative record, consisting of 327 laws and decrees, over the program's design (2018–21) and implementation (2022–25) phases. An LLM ensemble classifies every source against a fixed problem schema. We show that farmer-facing and state–donor sources represent implementation problems differently, and that the gap is systematic. Implementation laws reflect both donor and farmer information, but year-to-year legislative content covaries more strongly with donor content. The representation gap between state—donor—citizen is largest for local content on accountability and implementation failures. Thus, donor knowledge can fill an information vacuum at the design stage, but we find no evidence that it replaces the implementation information held by policy users.

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[†]Castañeda Dower: (email: pdower@wisc.edu)

# 1 Introduction

A tension exists between the international donor community's commitment to evidence-based policy and the necessity of working in nondemocratic countries to tackle the global challenges of security and climate.[1] Since democracy appears to be receding globally (Diamond, 2015), this tension should only amplify.

In nondemocratic countries, access to credible local information can be limited (Kenyon and Naoi, 2010), making it hard to gauge policy effectiveness. Policymakers in these countries face obstacles in obtaining unbiased information about local conditions (Pan and Chen, 2018): open channels risk surfacing sentiment that undermines the state's legitimacy (Guriev and Treisman, 2019), and citizens with a memory of repression may not speak truthfully. The same obstacles extend to researchers and to the international community. In Central Asia, for example, with its Soviet legacy of top-down policy making, information from users may not flow up to policymakers or to international stakeholders.

Donors and governments could, in principle, overcome these obstacles by substituting evidence of policy effectiveness from other contexts. That substitute is plausibly most useful at the design stage, when the relevant adoption barriers are generic and known from other settings. After rollout, the informational advantage shifts to policy users, who directly observe coercion, contractor behavior, equipment failure, water timing, and realized profitability. This paper asks whether donor knowledge can substitute for farmer feedback once a policy moves from design to implementation.

We study this question through Uzbekistan's flagship subsidy program for water-saving technologies (WST), which aims to address severe water scarcity in a formerly authoritarian regime. Informational sources from above and from below can produce double vision, whereby distinct images of the same policy emerge. The view from above, captured by official reports and state media, is likely to present the government's preferred narrative, emphasizing successes over implementation challenges. The view from below varies with local conditions and with the forum in which information is gathered.

As a benchmark, we apply the Kaleidoscope Model (KM). The KM posits that successful policy change involves moving through five stages (agenda setting, design, adoption, implementation, and evaluation). Success at each stage depends on determinants such as advocate support, veto players, focusing events, and available resources (Resnick et al., 2018; Haggblade et

[1] Authoritarian regimes are often perceived as able to implement climate policies rapidly (Beeson, 2010), and some argue that climate emergencies may legitimate coercive implementation (Mittiga, 2022), yet the resulting outcomes may be weaker than they could be (Gilley, 2012).

al., 2019).[2] Read from formal sources alone, the KM presents the WST subsidy as a policy success on target to meet its objective (Section 2). Uzbekistan is also a relatively favorable setting for policy learning. The government is reform-oriented, and drip irrigation is promoted as a win-win technology that saves water while raising yields. If the feedback loop fails here, it should fail more severely where incentives are less aligned.

To ask what happens to this reading when less constrained information is admitted, we compare five information channels that differ in their exposure to government influence, and the laws themselves: official government newspapers (Gov.News); the independent commercial news platform Kun.uz; focus group discussions and in-depth interviews (FGDs/IDIs); a pseudonymous Telegram chat of farmers; international donor and development-bank documents; and the problem-targeted laws and decrees themselves. The pseudonymity of the Telegram chat and the FGDs/IDIs that were arranged informally should represent the least constrained environment. We manually create a 14-problem schema, regrouped from a 19-category precursor (Appendix G), using data from 21 FGDs/IDIs of cotton farmers that we conducted in 15 different districts of Uzbekistan. LLM classifiers screen every corpus for WST relevance and assign the WST-positive texts their problems under the same schema. The first four channels capture how the policy is described. The last two let us measure how, and to whose description, formal policy responds.

Three findings emerge. First, there is an information wedge. The two farmer-facing forums overlap substantially in the problems that surface, while official sources cover a different and more generic mix; the pseudonymous chat most readily surfaces problems involving government control, such as coercion to adopt and contractor fraud, and the FGDs reveal that not all farmers who install drip irrigation use it. Second, the wedge is systematic. Donors and state media share one problem schema, farmer voice forms another, and the gap widens after rollout; within a problem, farmers, focus groups, and the press describe it in similar terms, while the law describes it in the terms donors use. Figure 1 previews the pattern: the problems for which the formal feedback loop works are largely those that are relatively easy for the central government to observe, while the problems where it breaks down concentrate among the hard-to-observe failures; the clearest exception is low yield/profit, which is observable but still goes unanswered. Third, the formal response is selective. Generic, donor-legible problems that belong to the global adoption template (such as cost, training, water supply shortage, and soil/climate mismatch) draw formal response; problems that surface only through local implementation (coercion, contractor fraud, faulty

[2]Policy change depends on decision-makers' knowledge of the problem (James and Jorgensen, 2009), on how actors perceive its costs and benefits (Vos et al., 2022), and, in the KM, on new information at the evaluation and reform stage (Resnick et al., 2018).

installation, corruption, equipment breakdown, water timing mismatch, and the profitability premise of the reform) are reflected far less readily. Implementation laws reflect both donor and farmer information, but year-to-year lawmaking covaries with donor attention while the farmer-voice association is not detectable in the main window and depends on timing; within the ten legislated problems (68 sub-aspects), the law's version of a problem tracks the donor version, with no detectable independent contribution from the farmer version once donor attention is held fixed. Forced adoption and low yield/profit, the two problems farmers raise most among those the law never touches, remain unanswered throughout implementation, and the laws that do exist are followed by no detectable decline in farmer complaints.

These findings speak directly to the political economy of development finance. Donor knowledge is an imperfect substitute for information on the ground, and development finance may orient formal attention toward donor-legible problems. This may bring short-run benefits but carries long-run costs for developing democratic institutions if key citizen problems go unaddressed. Substantively, we show that limited local information does more than conceal implementation failure: it produces a *selective* policy response. The evidence is consistent with donors and formal state channels substituting for weak or absent farmer feedback loops in legislative attention on generic, donor-legible problems. No such substitution is evident for the locally generated, politically costly failures that surface only after rollout.

We thereby contribute to work on citizen voice and government response in autocracies and to the theory of policy change in authoritarian contexts.[3] Pan and Chen (2018) documents an information wedge in officials' upward reports, which omit the corruption complaints that implicate them or their patrons; we show that a wedge exists even when incentives appear aligned. We differ on two margins from the literature on citizen expression as an information substitute in autocracies (Lorentzen, 2013, 2014; King et al., 2013; Qin et al., 2017): our outcome is legislative output rather than replies, enforcement actions, or reforms traced to particular complaints, and we ask whether international donors substitute for weak domestic feedback, a margin absent from these China studies (Chen et al., 2016; Distelhorst and Hou, 2017; Su and Meng, 2016; Jiang and Zeng, 2020; Buntaine et al., 2024). To our knowledge, the post-Soviet space is nearly absent from this work, even though Telegram carried citizen voice in Belarus's 2020 protests (Wijermars and Lokot, 2022) and Russian regional governments field complaints through VK (Zinnatullin, 2025). Climate policy also involves greater externalities and more complex problems than the programs

[3]Resnick et al. (2018) study policy change in Zambia, ranked 88th on the 2024 V-Dem electoral democracy index, relatively high for Sub-Saharan Africa and much higher than Uzbekistan, which ranks 144th (Coppedge et al., 2025).

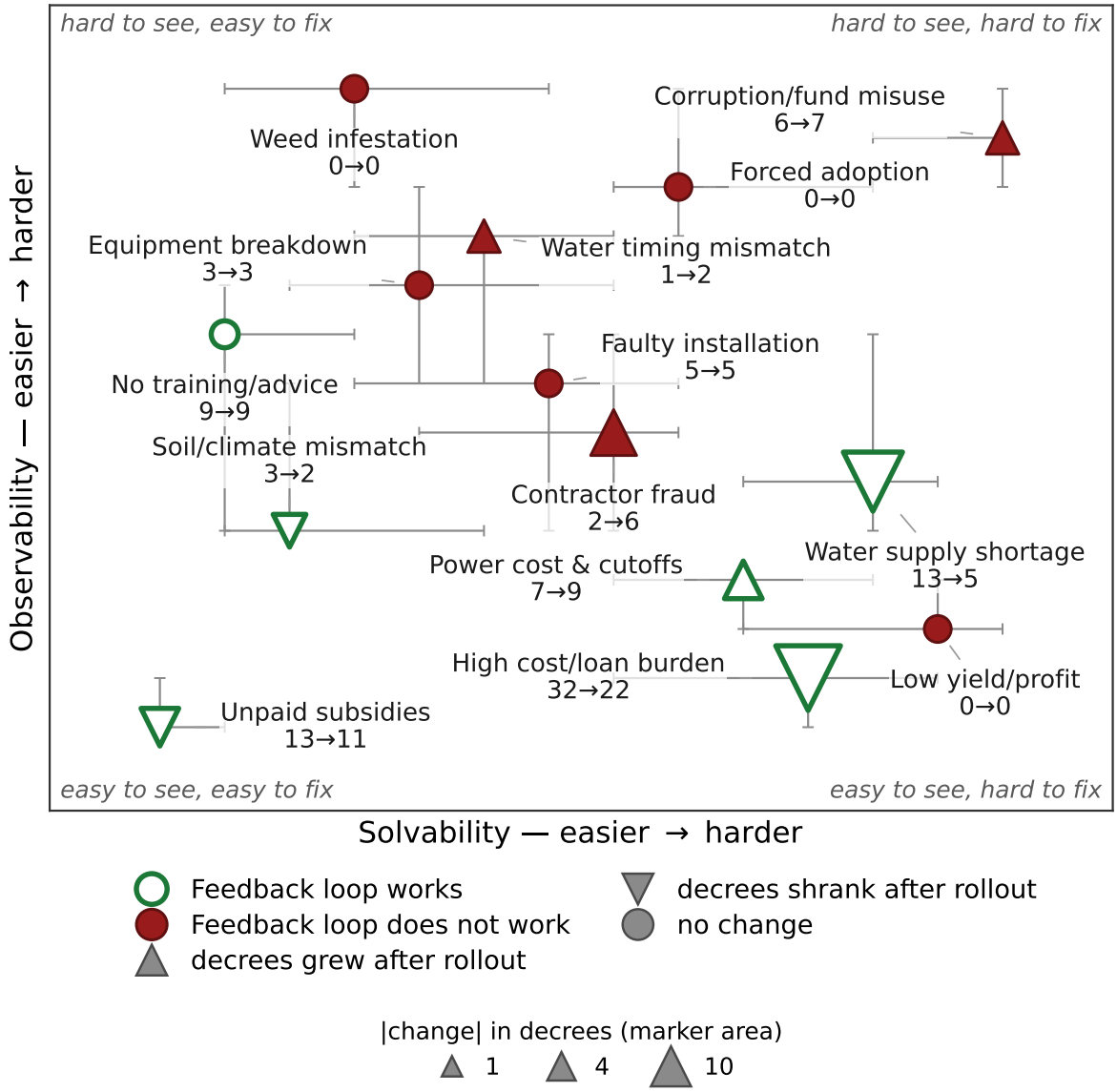



Figure 1 Preview of the main result. The 14 WST problems are placed by how observable they are to the central government and how solvable (both axes run from easier to harder, from expert pairwise ratings; whiskers are 80% credible intervals; Section 3). Open (white-filled) green markers mark problems whose formal feedback loop works; solid dark-red markers mark problems where it breaks down (the binary split of the five-profile typology of Section 4.4). Triangles show whether the problem's WST decree count grew (up) or shrank (down) after rollout, with marker area proportional to the change, and circles mark problems whose decree count did not change; labels carry the raw counts (deduplicated act basis, totals 94 → 81).

typically evaluated with the KM, which distinguishes the WST subsidy from fertilizer and other input subsidies.

Methodologically, each element of our design has precedence in the literature: LLM classification against a fixed schema (Fang et al., 2025), one codebook carried across text genres (Galiani et al., 2026), and multilingual automation of a shared codebook (Sebők et al., 2025); scoring differently positioned sources against a common reference is itself an established design (Gentzkow and Shapiro, 2010; Qin et al., 2018; Zhang et al., 2024), and coding legislation against a fixed topic schema descends from the Comparative Agendas Project (Jones and Baumgartner, 2005).[4] The conjunction, to our knowledge, does not. No existing study spans six institutionally independent genres under one canonical schema, benchmarked against a human-coded sample, and then uses the results as inputs to an analysis of policy response at a scale manual coding cannot reach (Section 3). Our six genres are citizen chat, interview transcripts, commercial media, state print media, donor documents, and legal text in two lower-resource languages.

# 2 Policy Context and Argument

## 2.1 The WST subsidy program

Uzbekistan is an arid, landlocked country whose water comes mainly from the Amu Darya and Syr Darya. Soviet-era irrigation built along these rivers to supply cotton contributed to severe environmental degradation, including the shrinking of the Aral Sea. Agriculture remains central: in 2023 it contributed about 20% of GDP and employed roughly a quarter of the labor force (World Bank, 2026); it consumes about 90% of the country's water through outdated infrastructure that loses water, and 56% of irrigated land depends on pumping (World Bank, 2025). Many farmers still irrigate by furrow, so water efficiency is a major policy concern.

Water-saving technologies refer to a menu of technologies that consume less water in production; the most important is drip irrigation. A replicated field trial in the Karshi Steppe found that drip irrigation reduced water use by over 50% while raising crop yield by roughly 30% (Djumaboev et al., 2019), so the international community sees its adoption as a win-win policy. Gains, however, can vary substantially with local conditions (Van der Kooij et al., 2013), and the investment is large. Installation costs run about 25 million UZS per hectare

[4] Aid-project descriptions have been machine-classified against a fixed taxonomy (Toetzke et al., 2022); the closest task analogues are Mellon et al. (2024), who code open-ended "most important issue" survey responses into a 50-category schema, and Ashwin et al. (2026), who scale interpretive coding of development interviews with supervised NLP.

(roughly $2,000/ha at 2024 exchange rates, used for all dollar conversions below). Laser leveling is an alternative technology that can improve water efficiency, especially for furrow and flood irrigation; while substantially cheaper, it is less effective and is not considered a substitute for drip irrigation in Uzbekistan. For the purposes of our study, we focus on policy change related to water-savings technologies since investments require significant buy-in at the local level and local conditions matter for the success of the policy.

The mobilization of interest in tackling water inefficiency surged under Shavkat Mirziyoyev, elected president after the death of Islam Karimov in 2016. The Ministry of Water Resources (MoWR) was established in 2018, separated from the Ministry of Agriculture (MoA), to implement unified national water policies. In 2019, a state subsidy program on water-saving technologies was adopted; IWMI reports that its recommendation to shift subsidies from energy to water-saving technologies informed the move.[5] The program aimed to cover 450,000 hectares by 2022, a target raised from an initial 250,000. The subsidy offsets the installation cost by 8 million UZS per hectare (about $650/ha). Published cost-benefit estimates put the unsubsidized payback period at roughly three years for cotton and under two for orchard crops (Niyazmetov and Rudenko, 2014); the subsidy shortens it further.[6]

A sequence of presidential acts then scaled the program up: UP-6024 (July 2020) set the 2030 concept, with 2 million hectares under WST, roughly half of the country's irrigated land, and 600,000 hectares under drip irrigation; PP-4919 (December 2020) launched the drip-irrigation subsidy at scale; PP-5005 (February 2021) revised the drip milestone upward to 822,000 hectares within a 1.1-million-hectare WST plan; and PP-144 (March 2022) adjusted a rollout already under way. Later acts extended rather than closed these targets (UP-27 of February 2023 and UP-158 of September 2023).[7] By September 2023, coverage had reportedly reached about 1 million hectares, up from 28,000 in 2018.[8]

Institutional incentives complicate this picture. Because of the large energy subsidies to irrigation systems inherited from the Soviet Union, the MoWR's main budget source is energy subsidies for canal pump stations rather than WST support. Ministry figures shared in our

[5]PP-4087 (27 December 2018), Cabinet of Ministers Resolutions No. 103 (6 February 2019) and No. 664 (12 August 2019), and PP-4499 (25 October 2019), full texts in our law corpus (Section 3.1); IWMI's account of its role in the subsidy shift.

[6]Namangan province at 2013 prices, investment over the annual benefit relative to furrow irrigation: 3.3 years for cotton, 4.3 for wheat, 1.7 for orchards (Niyazmetov and Rudenko, 2014). The subsidy did not yet exist.

[7]Presidential decrees UP-6024 (10 July 2020), UP-27 (28 February 2023) and UP-158 (11 September 2023), and presidential resolutions PP-4919 (11 December 2020), PP-5005 (24 February 2021) and PP-144 (1 March 2022); full texts in our law corpus (Section 3.1). The full chronology of the policy episode is summarized in Appendix Table 10.

[8]SUEZ, a private partner in the sector, "Water conservation and efficient usage in Uzbekistan", 22 September 2023; the same source reports 1.465 trillion UZS in subsidies over 2019–2022.

interviews put the disparity at roughly seventy to one in 2024. This creates a disincentive for the ministry to save energy, and to reduce the associated water extraction, at the farmer level, since WST could reduce energy use by reducing the need to pump water to distant irrigation areas. The ministry instead favors spending on concrete-lined canal systems, claiming that 40% of water loss is due to insufficient lining; since 2024 this preference has been carried into decreed budget lines (Section 5).[9]

Agricultural input subsidies are also associated with rent-seeking, and Uzbekistan is no exception. Key avenues likely include unofficial increases in interest rates on installment loans, unofficial delays or reductions in subsidy payouts, and unofficial charges for supplying consistent energy. More generally, institutional actors who would prefer the funds be redirected toward other areas may undermine the policy. We document farmer reports bearing on several of these margins in Section 4 and Appendix F.

When the episode is read through formal sources, the KM determinants nevertheless line up favorably at nearly every stage. Inefficient water use was recognized even under Karimov; his death and Mirziyoyev's election were a focusing event that catalyzed reform, and the enhanced Paris pledge at COP26 in 2021 reflects sustained momentum. Research inside and outside Uzbekistan celebrates the efficiency gains of WST while recognizing that benefits are heterogeneous, so local conditions matter: farmers need information about costs, including implementation and management costs, face uncertain benefits, and in some water-scarce areas severe shortages make WST impractical. This is the sole determinant coded negatively in the benchmark, the norms-and-beliefs row of Table 9. Beyond it, powerful advocates (IWMI, the donor community, the president himself) pushed adoption, no government veto player blocked the reform, and climate change would be expected to keep support growing.

This benchmark coding, stage by stage, appears in the first assessment column of Table 9, which Section 5 revisits with the view-from-below information. Figure 2 summarizes how information about WST performance flows among the main actors and where our channels and the legislative output sit in that flow (actor-by-actor positions in Appendix A).

## 2.2 Hypotheses

By double vision we do not mean merely different opinions about the same phenomenon: the informational sources must be distorted or misaligned enough to produce distinct characterizations of it. Sources differ in government control, oversight, and influence. The greater the government's involvement, the more accurately a source reflects the view from above; as involvement recedes, the source should better represent citizens' everyday experience. Donor

[9]Public reporting corroborates the 40% loss figure and water-sector spending of nearly $2 billion over 2022–23. We treat these interview-based figures as contextual observations rather than documented estimates.

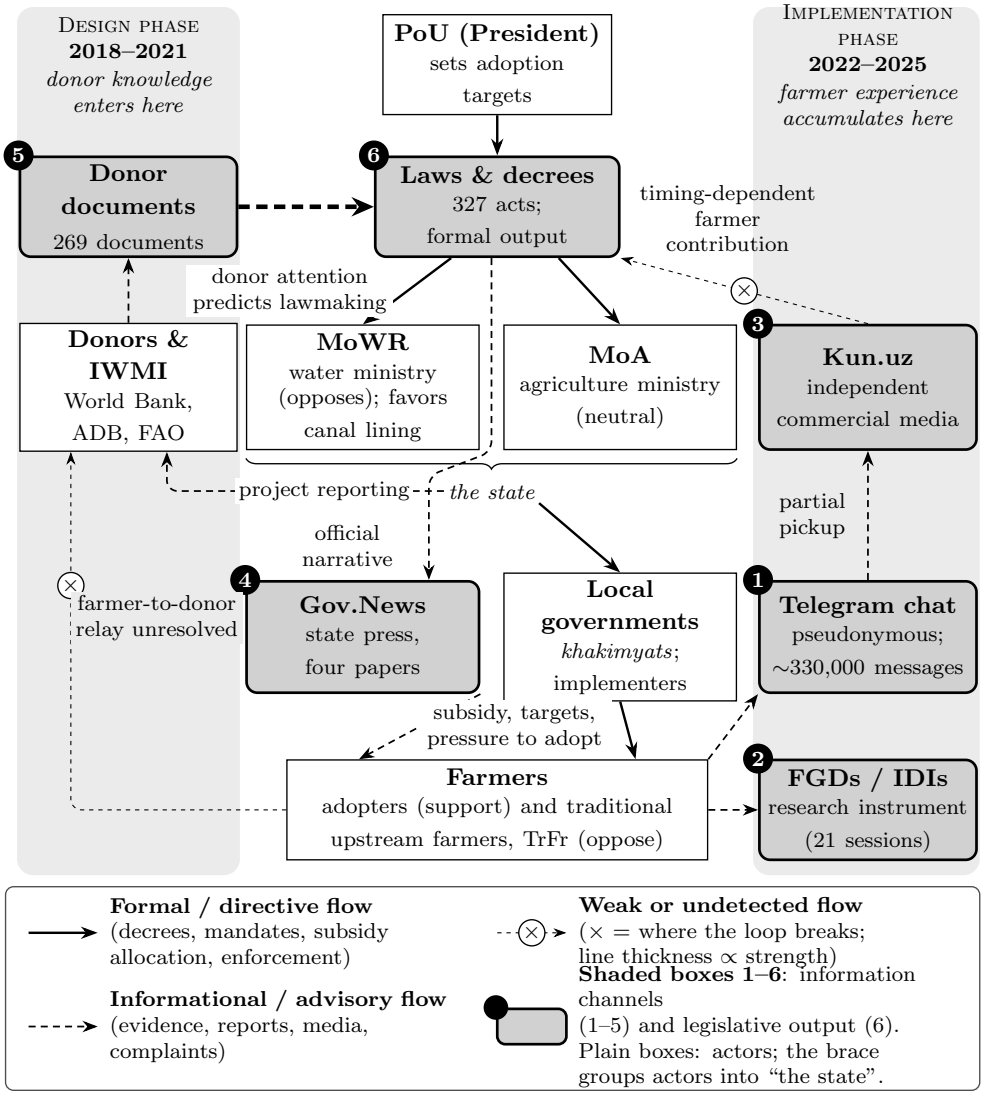



Figure 2 Information flows about WST performance among the main actors, and where the six numbered corpora of Table 1 sit in that flow (box 6 is the legislative output). Solid arrows are formal, directive flows; dashed arrows are informational or advisory flows; line thickness marks strength, and ⊗ marks where a flow thins or breaks (legend in figure); the official-narrative arrow marks Gov.News as the central state's press outlet. Donor knowledge enters mainly at design and reaches lawmaking through the donor-document channel. Farmer experience accumulates during implementation and surfaces in the pseudonymous Telegram chat and the FGDs/IDIs. Its contribution to formal lawmaking is timing-dependent, and whether donors relay farmer voice remains unresolved (Section 4.3). MoWR opposes farm-level WST and favors canal infrastructure; actor-by-actor stances are in Appendix A (Figure 8).

documents sit outside this domestic gradient: they are free of government control but reflect organizational portfolios rather than a survey of field conditions (Section 3), so where they fall between the two views is an empirical question. Systematic wedges between the two descriptions matter because external observers and domestic policymakers alike will struggle to update beliefs about policy effectiveness; such wedges are documented inside authoritarian bureaucracies (Pan and Chen, 2018), even though tolerated protest and watchdog reporting can supply an information-constrained government with feedback it otherwise lacks (Lorentzen, 2013, 2014). Establishing that a wedge exists does not, however, imply that it is systematic; it could reflect purely technological differences among sources.

*Hypothesis 1 (Information Wedge)*: The view from above and the view from below do not describe the same set of WST implementation problems. View-from-above sources (official reports, state-influenced media, and the problem-targeted laws themselves) cover a different mix of problems than farmer-facing sources and omit several locally revealed implementation failures that are prominent in the view from below. We read the laws' problem mentions as the formal system's own description. Donor documents enter the same coverage comparison; their alignment with the formal block is the subject of H2. For a win-win policy promoted as an unambiguous success, such a wedge is not expected on technological grounds alone.

*Hypothesis 2 (Systematic Wedge)*: The wedge is systematic rather than a random omission of a few topics. We hypothesize that the formal block (international donors and state media) shares one problem schema, while farmer-facing channels (Telegram and, during implementation, Kun.uz) share another, and that the divergence widens over the policy cycle as farmer voice rises. The laws' own problem mix is the legislative output examined under H3. If low-observability problems are disproportionately absent from the formal block, technological constraints are part of the story; if the absent problems are also the politically costly and harder-to-solve ones, and the formal block is internally aligned but orthogonal to farmer voice, then strategic sorting is also at work.

*Hypothesis 3 (Selective Policy Response)*: With a systematic wedge, we hypothesize that the policy response is selective. Throughout, the formal response we measure is problem-targeted legislative output (laws and decrees); administrative or budgetary responses outside lawmaking are not observed. Where problems are generic, donor-legible (Honig, 2018), and part of the global adoption template, international donors can substitute for weak domestic feedback, and formal response tracks the problem. Where problems are revealed only through local implementation and are politically costly or hard to verify, formal response is misaligned with the problems farmers report. Problems of this second kind are the seven the schema labels forced adoption, contractor fraud, faulty installation, corruption/fund misuse, equipment breakdown, water timing mismatch, and low yield/profit; the other seven

(high cost/loan burden, unpaid subsidies, no training/advice, power cost & cutoffs, water supply shortage, soil/climate mismatch, and weed infestation) are the generic problems. Donor substitution should be strongest in design, when the relevant barriers are generic and known from other WST settings. Farmer feedback should become more informative in implementation, when local enforcement, contractor quality, water timing, and realized profitability are revealed. The empirical question is whether formal channels update when that information advantage shifts from donors to farmers. The donor side of this mechanism is consistent with isomorphic mimicry (Pritchett et al., 2013): states adopt the policy forms that donors recognize and fund, so formal response can track the global template even where local feedback is weak. If response is present for donor- and state-legible problems but muted exactly where farmers hold the information advantage, then policy response is selectively, not uniformly, misaligned. A closing loop also carries a dynamic implication: if the formal response resolves a problem, farmer complaints about it should eventually decline.

# 3 Data and Measurement

We assemble five information channels and the legislative record. Every text is first passed through a binary screen for water-saving technology; texts that pass are WST-positive and are the units classified against the 14-problem schema (Section 3.2). Table 1 summarizes each corpus, its WST-positive count, its analytical role, and its principal limitation.

## 3.1 Sources

**Design vs. implementation phases.** We split all series into a design phase (2018–2021), anchored to presidential resolution PP-4919 of December 2020, which launched the drip-irrigation subsidy at scale, and an implementation phase (2022–2025). Although PP-4919 dates to December 2020, we keep 2021 in the design phase: PP-5005 (February 2021) was still revising the drip milestones upward, and PP-144 (March 2022) adjusts a rollout already under way. The split is substantive for the reasons argued in Section 2.2: donor substitution should be strongest at design, farmer feedback most informative during implementation. Channel coverage is asymmetric at design (Gov.News digitization begins in 2019 and Telegram begins in late 2020), so design-phase rates for those channels rest on partial windows.

**Farmer-facing sources.** To gain insight into farmers' perceptions, we employ two complementary sources: messages from a large farmer-oriented Telegram chat and Focus Group Discussions

Table 1 The five information channels and the legislative-output corpus. WST-positive texts are those that pass the binary screen of Section 3.2; they are the units classified against the 14-problem schema. All analyses end on 31 December 2025; records dated 2026 are excluded. Ministry interviews cited in Section 2 provide context only and are not classified.

| **Corpus** | **Years** | **Unit and $N$** | **Analytical role** | **Principal limitation** |
|---|---|---|---|---|
| Telegram chat | 2020–25 | 328,076 messages (5,685 WST-positive) | Continuous farmer-facing signal; enters all time-series and substitution tests | Self-selected participants |
| FGDs/IDIs | 2023 | 21 sessions, 15 districts (21 transcripts, all WST-positive) | Validate chat problems; context on actual use | One-time, small sample |
| Kun.uz | 2015–25 | 127,771 articles (470 WST-positive) | Independent commercial media; the commercial middle | Market-driven selection; sparse WST coverage |
| Gov.News | 2019–25 | ∼146,000 OCR-derived articles, 4 papers (3,219 WST-positive) | State media; the view from above | State control of content |
| Donor documents | 2001–25 | 269 documents (12 organizations) | Donor attention; substitution regressor | Reflects portfolios; delayed evaluations |
| Laws and decrees | 1999–2025 | 327 distinct acts (462 bilingual records) | Formal output; outcome variable | Administrative and budgetary responses not observed |

(FGDs) with In-Depth Interviews (IDIs).[10] The two play different roles. Telegram is the workhorse: only it provides the continuous, high-volume signal that the time-series, alignment, and substitution analyses require. The dataset contains 328,076 messages from a dedicated agricultural chat group, December 2020 through December 2025, of which 5,685 pass the WST screen (5,495 of them dated in the 2022–25 implementation phase). Sender metadata identify approximately 20,500 unique posting accounts, but they do not establish the chat's total membership or verify farmer status. The chat continues into 2026; we stop at the end of 2025, the common cutoff applied to every channel.

The FGDs/IDIs are a one-time, systematically conducted, small-sample instrument: we use them to validate that the problems surfacing in the chat are real and to add context on whether installed systems are actually used. They covered 15 districts across 5 regions, with data collection over a two-month period in 2023; 21 sessions were held with 1–5 farmers each. Recruitment initially went through the local government (*khakimyat*) and then switched to wider nets, including personal networks, cold requests at the farm gate, and agricultural cluster networks. We expect the least constrained environments to be discussions carried out under pseudonymous handles, as on Telegram, and FGDs arranged informally; Section 4.1 reports how the two recruitment modes differed. Neither farmer-facing source is representative of farmers as a whole, and both remain subject to self-selection. All sessions were recorded and transcribed, and the transcripts are classified with the same ensemble and schema as every other channel.

**Independent and state media.** Kun.uz is Uzbekistan's leading digital news platform: it reports approximately 350,000 daily visitors (Kun.uz, 2026), is the news source most cited on Uzbek social media (Kun.uz, 2025; Kursiv, 2026), and publishes mainly in Uzbek with Russian and English editions.[11] Our corpus comprises 127,771 articles published 2015–2025, of which 470 are WST-positive; they are classified with the same ensemble and schema as every other channel. Analytically, Kun.uz occupies the commercial middle between farmer voice and the state, since it is market-driven rather than state-directed.

The official print press forms the opposite pole; this is the view from above in its purest form. Four state-affiliated newspapers (*Pravda Vostoka*, *Narodnoe Slovo*, *Yangi O'zbekiston*, and *Ishonch*/*Doverie*) remain tightly controlled, and the state can use them to shape the narrative about the subsidy program; we would expect the program to be characterized as an overwhelming success. Their WST coverage is selective in a second sense: limited

[10] We conducted these FGDs and IDIs with farmers to prepare for a large-scale survey and impact evaluation of the subsidy program.

[11] The visitor figure is the outlet's own Google Analytics count; an independent estimator puts traffic at roughly 200,000 visits a day, excluding the app and Telegram audiences (Similarweb, 2026).

space and competing strategic messaging may crowd out articles that would otherwise have run. Because these papers circulate only as scanned print issues, we digitized them with a custom OCR pipeline (DocLayout-YOLO segmentation, Surya OCR) over 2019–2025, yielding roughly 146,000 article-level texts pooled into the *Gov.News* channel.[12]

**Donor documents and laws.** The donor documents and the laws capture the view from the international development community and the formal policy response itself: respectively the explanatory channel and the outcome variable of the substitution analysis (Section 4.3). The *donor corpus* is harvested from 16 international donor and development organizations; after deduplication and a narrow WST-relevance screen, it comprises 269 documents across 12 organizations published through 2025 (five further documents dated 2026 are excluded). The four with substantial WST coverage (the World Bank, the Asian Development Bank, FAO, and IWMI) contribute 255 of the 269; the remaining eight organizations contribute too few WST-relevant documents for organization-level comparisons.[13] Donor coverage of a problem in a phase is the share of that phase's documents that mention it. Documents are dated by publication year and assigned to the same phases as the other channels (design-phase documents $n = 96$, implementation $n = 111$).[14] Donor publication is itself selective: documents reflect organizational priorities and project portfolios rather than a survey of field conditions.

The *law corpus* consists of 327 distinct WST-positive laws and decrees enacted through 2025, collected as 462 bilingual records and collapsed into single acts by act type, document number, and enactment date; 12 further acts dated 2026 are excluded. The bilingual twin pairs double as a free test–retest reliability check on the classifier. All law counts in the paper are on this distinct-act basis. Tagged acts yield 185 through-2025 act×problem links: 175 fall in the two analytical phases (94 design-phase, 81 implementation-phase), and 10 predate 2018.

**Ethics and data handling.** Members of the Telegram group participate under self-chosen handles. We analyze messages in aggregate, report no usernames or other identifying details, and present only short translated excerpts (Appendix F). The FGD/IDI sessions were recorded and transcribed, and no participant names are reported.

[12] Article counts by paper: *Yangi O'zbekiston* 47,537, *Narodnoe Slovo* 46,346, *Pravda Vostoka* 32,927, and *Ishonch/Doverie* 18,898.

[13] The 16 organizations screened span multilateral development banks (World Bank, ADB, AIIB, EBRD, IFAD), UN technical agencies (FAO, UNDP), CGIAR research institutes (IWMI, ICARDA), bilateral and supranational agencies (GIZ, JICA, SDC, GGGI, EU), and vertical climate funds (GEF, GCF).

[14] The remaining 62 documents, dated 2001–2017, enter no phase-level rate, timeline, or panel; they contribute only to the sub-aspect comparison of Section 4.3, which pools all donor documents.

## 3.2 Measurement: fourteen problems and 94 sub-aspects

All sources are classified against a single canonical schema of fourteen WST problems, so that coverage is directly comparable across channels. Classification proceeds in two steps. First, a binary screen establishes whether a text concerns WST; it yields 9,395 WST-positive texts across the four domestic channels, among them 5,685 Telegram messages and 470 Kun.uz articles. Donor documents and laws are screened in separate pipelines. Second, each WST-positive domestic text is assigned its relevant problems by a three-model ensemble of two locally run open-weight models with a larger cloud model breaking disagreements, under a per-problem majority vote (2 of 3). Roughly 90% of screened texts are resolved by agreement of the two local models, most of them texts both leave untagged; among texts in which either local model finds a problem, about a third are resolved locally and the rest go to the cloud tiebreaker.[15] The law and donor corpora are assigned problems by the same three-model ensemble and 2-of-3 vote, with corpus-specific prompt framing (for laws, the problems an act is designed to address through concrete measures) and longer-document chunking; the corresponding prompts are likewise in the supplementary file.

The canonical schema comprises fourteen WST problems grouped under eight categories, applied identically across all six corpora:

**Equipment & Installation:** Equipment breakdown (drip lines, filters, pumps breaking or clogging); Faulty installation (poor quality, incorrect setup, contractor errors)

**Access & Affordability:** High cost/loan burden (equipment, installation, maintenance); Unpaid subsidies (delayed, partial, or denied payments)

**Trust & Accountability:** Contractor fraud; Corruption/fund misuse

**Infrastructure & Operations:** Power cost & cutoffs; Water supply shortage; Water timing mismatch

**Governance & Policy:** Forced adoption (coercion into WST contracts/loans under administrative pressure)

**Knowledge & Support:** No training/advice (knowledge and technical guidance to operate and maintain WST)

[15] The screen uses a local Gemma-4 (31B) classifier. The ensemble pairs Qwen3.5 (27B) and Gemma-4 (31B) locally, with Kimi K2.5 (via the Moonshot API and OpenRouter) as tiebreaker, with a recheck of tags receiving exactly one of three votes by a single second model (Gemma-4 or Kimi K2.5, whichever did not cast the vote), whose confirmation adds the tag. Among 76 Kun.uz articles recovered after the production run, two labels received author-approved human corrections following independent Qwen3.8, Luna, and Sol review: one contractor-fraud tag was removed and one equipment-breakdown tag added; the original model votes remain preserved. The full production prompts and configuration for every classifier stage are reproduced in the supplementary prompts file.

**Technical Limitations:** Soil/climate mismatch (saline soils, high groundwater, heavy clay, extreme heat); Weed infestation

**Outcome Failure:** Low yield/profit

**Sub-aspects.** Below the problem level, each of the fourteen problems is decomposed into five to eight concrete sub-aspects, 94 in all, so that we can ask not only which problems a source names but how it describes each one (Appendix E). Every donor document and law tagged with a problem, the implementation-phase Telegram messages tagged with it (up to 200 per problem, a cap that binds for high cost/loan burden and unpaid subsidies), and the implementation-phase Gov.News, Kun.uz, and FGD texts tagged with it are classified in full against that problem's sub-aspect codebook by two independent coders, Qwen3.8 and GPT-5.6 Luna; the primary label is the union of the two, and the model-specific labels serve as robustness checks. A source's version of a problem is its distribution of assignments across the problem's sub-aspects.

**Validation.** A domain expert independently classified a 100-text sample, 87 of them Telegram messages, while blind to the models (Gilardi et al., 2023; Mellon et al., 2024). On the binary task (does a text receive at least one of the 14 problem labels?), detection matches the human coder closely (accuracy 0.86, $F_1 \approx 0.90$). Agreement on *which* problems a text raises is moderate (multi-label $F_1 \approx 0.57$). The decision rule favors precision over recall: a tag needs two of three votes, and a single-vote tag survives only if a second model confirms it in a targeted recheck,[16] so the classifier under-reports rather than fabricates problems. On Telegram, the one channel where the sample is large enough to check, the ensemble reproduces the coder's problem-share ranking almost exactly (Spearman $\rho = 0.90$ across the crosswalked problem buckets; Appendix G). We therefore treat problem-level counts as measured with error and rest every inference on relative rankings, schema alignment, and within-problem comparisons across sources rather than on absolute prevalence (Egami et al., 2023; Ludwig et al., 2025). The sample's composition, the differential-error test across sources, the single-coder and precursor-schema caveats, and the coder agreement at the sub-aspect level are reported in Appendix G.

## 3.3 Hypothesis tests

For inference, we follow the process-tracing logic of Mahoney (2012) and Collier (2011). Each hypothesis is first put to a hoop test, a necessary-condition check: failing it eliminates the

[16] In production, about 250 of 1,192 single-vote tags on the domestic corpora and 29 of 202 on the law records were confirmed.

hypothesis, while passing keeps it alive without confirming it. For that logic to bind, the failing observation has to be named in advance, so for every step of the argument we state the observation that would have counted against it (Table 2), and where the data permit we add a statistical test whose null is that observation. Table 2 collects the tests, the alternatives they address, and those failing observations.

**H1: View from above and view from below cover different problems.** For every channel and phase, a problem's coverage rate is the share of that channel's WST-positive texts that mention it (for donors, of documents; for laws, of act–problem links). H1 fails if the farmer-facing channels and the official channels name the same problems at similar relative rates. The comparison concerns both margins, which problems appear at all and how often. Additionally, we perform a sentiment comparison to see whether the problems absent from the view from above also attract more negative reports from below.

**H2: The information wedge is systematic.** At the problem level we compute Spearman rank correlations of coverage across the 14 problems for every pair of channels, separately by phase; with $n = 14$, $|\rho| \geq 0.53$ is required for $p < 0.05$, and differences between two correlations that share a channel are tested with Williams's test (Williams, 1959) and a bootstrap (Appendix D). H2 fails if the donor and state channels are no more aligned with one another than with Telegram. Attention timelines show the same rates by year (problem tags per 100 WST-positive texts for the domestic channels, per 100 documents for donors, and act–problem links for laws); whether the divergence widens after rollout is reported descriptively, since the design-phase farmer signal rests on too few messages for a cross-phase test. Below the problem level, we ask whether two sources describe a given problem in the same terms: for each pair of sources, sub-aspect shares are demeaned within each problem and correlated across the 94 sub-aspects, with inference by permuting one source's shares within problems. The block structure of H2 fails at this level if the farmer accounts are no more aligned with one another than with the donor account.

**H3: Lawmaking does not follow view from below in the implementation phase.** Two tests carry H3. The first works below the problem level. The unit is a sub-aspect within a problem; the outcome is the number of distinct law acts carrying it; the regressors are the sub-aspect's share of donor assignments and of Telegram assignments within its problem; problem fixed effects absorb which problems receive laws at all, so the coefficients compare versions of the same problem. Estimation is Poisson; $p$-values permute one regressor across sub-aspects within each problem (5,000 draws), and the donor minus farmer difference is

tested by swapping the two shares sub-aspect by sub-aspect. H3 fails if the farmer share predicts law counts as well as the donor share. Selectivity is tested directly by letting both slopes differ between the six generic problems that reach law, the shared problems of Table 7 (weed infestation, the seventh generic problem, has no law act), and the locally revealed problems: H3's selective form fails if the donor slope is no smaller on the locally revealed problems. The second test works at the problem level over time: a problem×year Poisson panel of law counts on the previous year's donor attention and Telegram attention, each a channel's share of problem mentions in that year, with year fixed effects, standard errors clustered by problem, and within-year permutation inference; the main window is 2021–2025 (70 problem-years, since lagged farmer attention exists only from the chat's start in late 2020), with 2023–2025 and a two-year lag as timing variants. The panel follows the agenda-setting design of Guinaudeau and Guinaudeau (2023), with donor-document and farmer-chat attention in place of manifesto salience; decree numbers reach the chat within days for major laws, so farmers observe the legislative output. The panel's H3 fails if lagged farmer attention predicts next-year law counts as well as donor attention does. Two companions bear on substitution rather than association: a phase-level model comparison asks whether adding farmer attention to a donor-only model of implementation-phase law counts improves fit, and vice versa; and a transmission check regresses donor attention on lagged farmer attention, since substitution in the strong sense requires that donors do not simply relay farmer complaints. Per-problem Jensen–Shannon contrasts of the law's distance from the donor and farmer accounts complete the sub-aspect evidence (Appendix E).

## 3.4 Mechanism diagnostics and robustness

In this subsection, we probe why uptake differs and whether the pattern is specific to this policy. First, an expert rater ranked the 14 problems on how observable each is to the central government, how solvable it is, and how sensitive it is to donor pressure, through pairwise comparisons fit with a Bradley–Terry model (Carlson and Montgomery, 2017; Davidson, 1970); the rater is one of the authors, the credible intervals describe the one-rater fit only, and archived LLM rating sessions reproduce the rankings (Appendix C). Second, a policy-learning typology clusters the 14 problems on their profile of design- and implementation-phase attention across four channels (Telegram, Kun.uz, Gov.News, and donors) and the law (15 signals, two phase rates and their shift per source; cosine distance; five profiles); it summarizes the joint pattern rather than testing it, and its stability is reported in Appendix B. Third, a separate classifier assigns sentiment to each implementation-phase Telegram and Gov.News text and, for each text–problem pair, whom it blames: state actors,

private actors, the system, or no one (Appendix A). Fourth, three checks on the response side. A weekly event study around exact enactment dates asks whether farmer complaints about a problem fall after a law addresses it, using the 63 implementation-phase act–problem links with complete ±26-week Telegram windows, with the announcement-week dip, decree-discussion threads, seasonality, and a self-calibrated null as diagnostics (Hobbs and Roberts, 2018). An observer chain asks whether donor attention that predicts lawmaking also predicts fewer negative farmer reports the following year. A topic placebo repeats the Telegram–Gov.News comparison for three topics with no policy loop of their own, adverse weather, pests and crop disease, and the consequences of the Russia–Ukraine war, to check that the wedge is specific to WST rather than a generic difference between a farmer chat and a state newspaper (Appendix E).

# 4 Main Results

Table 2 maps each step of the argument to the evidence that carries it.

## 4.1 H1: An Information Wedge Exists

Figure 3 shows the extensive and intensive margins of the information wedge. For reference, it adds donor documents and WST laws as rows alongside the public and official media, so the contrast is not only farmer voice versus Gov.News. Some problems are visible across all channels: high cost/loan burden and unpaid subsidies. Others are concentrated in Telegram and Kun.uz, or nearly absent from official coverage and legal response.

For example, *forced adoption* appears in 2.7% of WST-positive Telegram posts and 1.6% of Kun.uz articles, but in under 0.05% of Gov.News articles and in no implementation-phase law. *Contractor fraud* is similarly public-facing: 2.9% in Telegram and 2.2% in Kun.uz, compared with 0.18% in Gov.News and none in donor documents.[17]

In the FGDs/IDIs, most farmers raise serious concerns with the program's implementation, though some report positive views of drip irrigation and the subsidy program. Unlike furrow irrigation, drip systems require a local power source to maintain pressure. Connecting to the grid and weathering outages impose costs the program does not compensate. Farmers also report little choice among installation companies, some of which use poor-quality materials. No single problem is universal, but high cost (installation and loan burden) appears in 8 of 21 discussions (38%). Among the 11 sessions that include farmers with installed or purchased

[17] FGDs broadly validate the Telegram ordering despite their smaller sample: they recover most of the same problem families while adding contextual detail on why farmers do or do not use installed WST systems. FGDs do not enter the time-series analyses that follow (Section 3.1).

Table 2 How the evidence accumulates: from an information wedge to selective policy response. Each row links one claim to the evidence that most directly supports it and to the main alternative interpretation that evidence addresses, and states the observation that would have counted against it. The claims are cumulative, building to the KM reassessment in Section 5; no single diagnostic is decisive on its own.

| Step | Claim | Evidence | Main alternative addressed | Observation that would have failed it |
|---|---|---|---|---|
| **1. Wedge exists** (H1) | The same 14 problems appear with different emphasis across forums. | Coverage heatmap (Fig. 3). | Shared problem mix. | Farmer-facing and official channels name the same problems at similar relative rates. |
| **2. Wedge is systematic** (H2) | Donor and state channels share one schema, farmer-facing channels another; the gap widens during implementation. Below the problem level, the farmer accounts cohere and the law's version of a problem follows the donors'. | Attention timeline (Fig. 4); cross-channel correlations (Table 4, Fig. 9); sub-aspect alignment (Table 3). | Idiosyncratic omission. | Donor and state channels no more aligned with one another than with Telegram; within problems, farmer accounts no more aligned with one another than with the donor account. |
| **3. Uptake is donor-associated and selective** (H3) | Donors and laws carry the pre-rollout formal agenda into implementation, and donor attention runs opposite to farmer attention on the locally revealed problems (a descriptive subset pattern). Legislative attention covaries with donor attention, while the farmer-voice association is not detectable in the main window and depends on timing; whether donors relay farmer voice is unresolved. Within a problem, the version that enters law follows the donor version: sub-aspect donor share predicts law counts, farmer share does not, on the shared problems especially. | Panel (Table 6), phase, provenance, observer-chain, and sub-aspect tests (Table 7, Section 4.3). | Complete or functionally adequate substitution. | Farmer sub-aspect shares predict law counts as well as donor shares; lagged farmer attention predicts next-year law counts as well as donor attention; the donor slope no smaller on locally revealed problems. |
| **4. Where uptake weakens** | Formal channels answer generic, donor-legible problems; local implementation failures far less readily. | Five-profile typology (Fig. 5). | Uniform state silence. | Weak-uptake problems spread evenly across profiles rather than concentrating among the locally revealed ones. |
| **5. Why uptake differs** | Breakdowns concentrate where problems are hard to observe centrally or accountability-entangled. | Observability and solvability ratings (Fig. 6); blame (Table 8). | Failure unrelated to legibility. | Broken-loop problems rated no less observable and no more accountability-entangled than working-loop ones. |
| **6. Boundary and robustness** | The wedge is WST-specific; no large post-law decline is detectable in farmer reports. | Topic placebo (Fig. 7); post-law event study. | Source style; generic grievance. | The placebo topics show the same Telegram–Gov.News divergence; complaints fall after laws. |
| **7. KM reassessment** | With the view from below, the KM success reading becomes mixed. | KM reevaluation (Table 9). | Uniform success verdict. | The view-from-below evidence leaves every determinant's benchmark coding unchanged. |

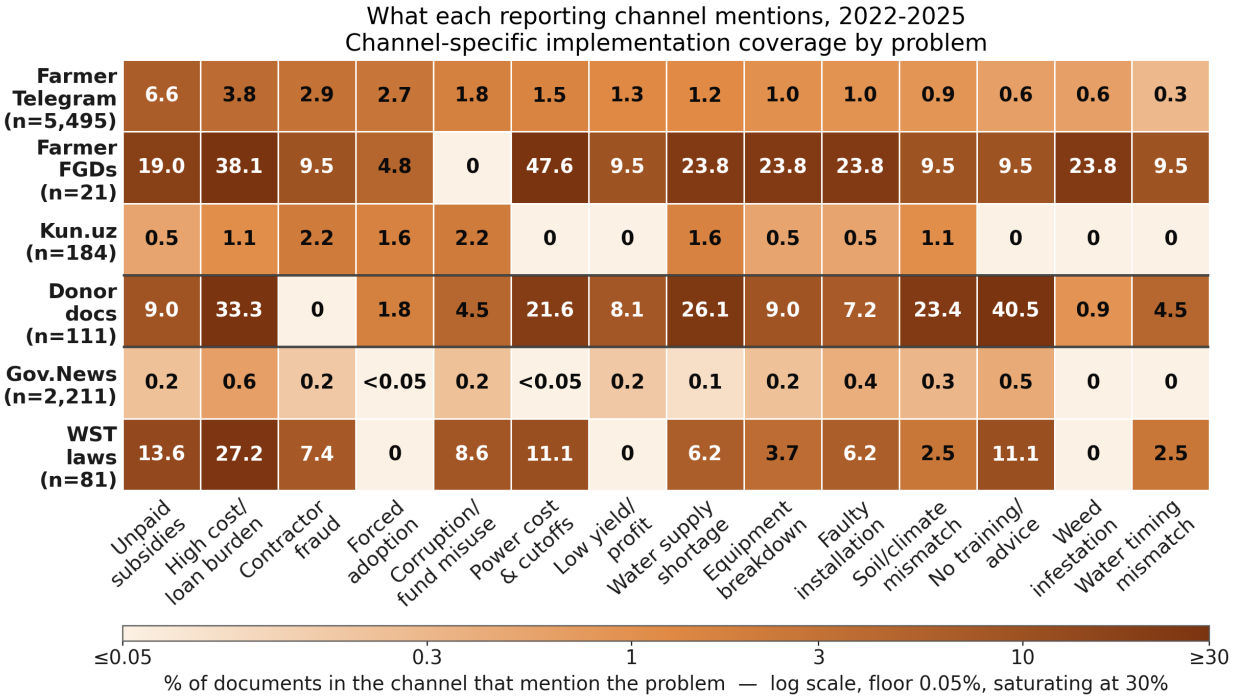



Figure 3 Per-problem coverage by channel, implementation phase 2022–25. Cells: percent of WST-positive texts (Telegram, FGDs, Kun.uz, Gov.News), percent of implementation-phase donor documents (donors; $n = 111$), and share of 81 implementation-phase distinct-act problem links (laws); per-channel $N$s are in the row labels. Columns are sorted by Telegram mention rate, and color follows the percentage scale shown below the matrix. The view from above and the view from below do not cover the same problems (H1).

systems, four describe systems idle or non-functioning ("Yes, we have drip irrigation, but there's no water"), and two further sessions report installed equipment breaking down within a year.

Recruitment mode shaped what farmers said. FGDs organized in coordination with the *khakimyat* surfaced fewer and more generic problems, typically framed around cost. Those arranged informally more readily raised institutional and service-related complaints, as Section 3.1 anticipated.

The Telegram chat raises much the same set of problems as the FGDs, and we read that overlap as evidence the problems are real. Contractor fraud and unpaid subsidies are among the most frequent problems in the chat, and neither is within farmers' control. One message is representative: *"Chinese contractor promised drip irrigation... deceived for 500 million."* Further quotations are collected in Appendix F.

Telegram and the FGDs identify more complaints than official sources, and the problem mix differs. Institutional and implementation failures are muted in official sources, which concentrate on the generic, program-template problems. H1 thereby passes the hoop test, and we turn to whether the wedge is systematic.

## 4.2 H2: The Wedge Is Systematic

Figure 4 shows the wedge opening over time. While the program was being designed, WST attention belonged to donors, the state press, and lawmakers. Implementation reversed the pattern. From 2022, farmer complaints on Telegram rose year on year while official coverage fell steeply, so that by 2025 Telegram was raising problems at roughly 36 times the rate of Gov.News. Lawmaking did not follow the farmers. Problem-targeted decrees peaked in 2022 and fell together with official coverage, not with farmer voice; the series partially rebounds by 2025 (to 24 links, from a trough of 6), still below the 2022 peak.

Kun.uz kept covering the farmers' problems after 2022 but covered them far less often. Its WST problem rate drops toward Gov.News levels, while its implementation-phase problem mix stays aligned with Telegram ($\rho = +0.55$, Table 4). A falling rate with a stable mix is consistent with supply-side filtering in a tighter media environment. The time series does not by itself identify censorship, and demand-side selection may also matter. The Kun.uz series does establish that the implementation-phase wedge is not only a Gov.News artifact.

The problem schemas tell the same story in cross-section. Ask which of the 14 problems each channel emphasizes, and the donor agencies and Gov.News give nearly the same answer: the donor groups' rankings correlate at $\rho = +0.64$ to $+0.92$ with one another, Gov.News tracks the donors during implementation, and design-phase laws lean the same way. Telegram gives

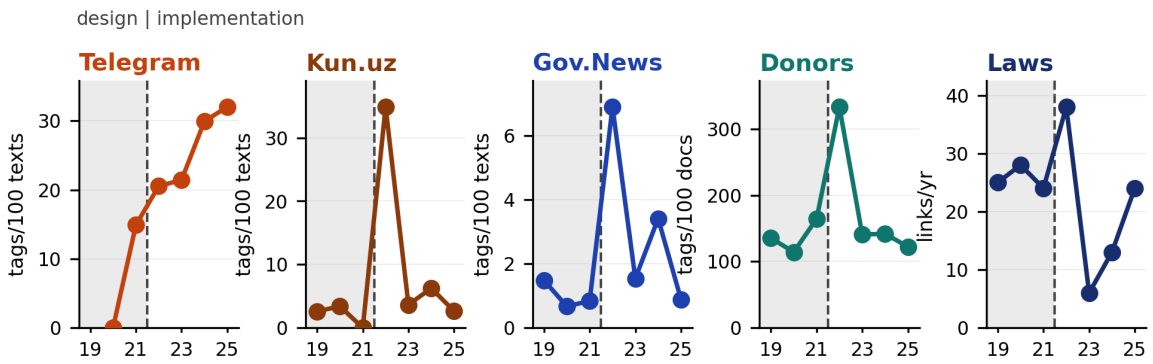



Figure 4 Aggregate WST attention by year for four information channels and the legislative record, 2019–2025. Units are channel-specific: problem tags per 100 WST-positive texts for Telegram, Kun.uz, and Gov.News; problem tags per 100 documents for donors; and problem–decree links per year for laws. Texts can carry several tags, so the first four series are tag intensities rather than shares of texts; $y$-scales are channel-specific and not comparable across panels. Farmer voice rises late, while Gov.News, donors, Kun.uz, and laws peak around the implementation launch and then fall (for laws, to a trough of 6 links before rebounding to 24 by 2025). Panels begin in 2019; the design phase itself starts in 2018, and its law links are included in the phase totals.

a different answer. During implementation its ranking is essentially uncorrelated with every donor channel ($-0.20$ to $+0.15$), and the one channel it agrees with is Kun.uz ($+0.55$). With 14 problems these correlations are noisy, yet the gap between the blocks is large enough to be statistically distinguishable in the strongest pairings: three of six contrasts in the exploratory donor-coherence panel pass the pre-set joint criterion, two surviving a Bonferroni correction, both in the implementation phase, where the Telegram profile rests on 1,035 problem-bearing messages rather than the 25 of the design phase (Table 4; Appendix D).

At the problem level, law content correlates similarly with Telegram and with the donor bundle ($\rho \approx +0.50$ each), so these headline correlations do not rank the sources. Section 4.3 shows that the two agreements rest on different problems and, below the problem level, on different formulations.

The same question can be asked below the problem level. Section 4.3 decomposes every problem into sub-aspects, and the resulting version shares let us ask whether two sources describe a given problem in the same terms, holding the problem fixed (Table 3). Across the 94 sub-aspects (fewer for pairs in which one source has no assignment; Table 3), the farmer channels agree with each other most closely, and both agree with the commercial and the state press. Donor documents overlap with the farmer accounts only moderately. The law is the outlier: its version of a problem correlates with the donor version and not at all with the farmer version, a difference that is statistically clear, while the state press sits between

the two. Below the problem level, then, the wedge separates the legislative text from the domestic discourse, with donor documents as the account the law follows most.

Table 3 Do sources describe a problem in the same terms? Within-problem correlations of sub-aspect assignment shares across the 94 sub-aspects (shares demeaned within each problem and pooled; union labels). Permutation $p$-values shuffle the second source's shares within each problem (5,000 draws).

| Source pair | Problems | Sub-aspects | Within-problem $r$ ($p$) |
|---|---|---|---|
| Telegram with FGDs | 13 | 88 | +0.53 ($< 0.001$) |
| Telegram with Kun.uz | 7 | 48 | +0.47 ($< 0.001$) |
| Telegram with Gov.News | 12 | 81 | +0.33 (0.005) |
| Telegram with donors | 14 | 94 | +0.26 (0.011) |
| Gov.News with donors | 12 | 81 | +0.26 (0.081) |
| Laws with donors | 10 | 68 | +0.56 ($< 0.001$) |
| Laws with Gov.News | 10 | 68 | +0.32 (0.020) |
| Laws with FGDs | 9 | 62 | +0.12 (0.34) |
| Laws with Telegram | 10 | 68 | +0.05 (0.71) |

Notes: The law–donor correlation exceeds the law–Telegram and law–FGD correlations (swap tests $p = 0.025$ and $p = 0.010$) but not separably the law–Gov.News correlation ($p = 0.14$). Gov.News is no closer to donors than to Telegram ($p = 0.76$). Luna-only labels reproduce every sign; Qwen-primary labels are flatter.

Table 4 Key problem-schema alignments. Cells are Spearman correlations across the 14 canonical problems; $|\rho| \geq 0.53$ is required for $p < 0.05$ at $n = 14$. The correlation matrices for the donor, Telegram, and law anchors are in Appendix A; the Gov.News correlations are computed identically but not displayed there; difference tests in Appendix D.

| Channel pair (phase) | $\rho$ |
|---|---|
| Donor institutions with each other (within and across phases) | +0.64 to +0.92 |
| Gov.News with three donor groups (implementation) | +0.53 to +0.58 |
| Gov.News with Telegram (implementation) | +0.21 |
| Design laws with Gov.News/three donor groups | +0.34 to +0.62 |
| Telegram with donor channels (implementation) | −0.20 to +0.15 |
| Telegram with Kun.uz (implementation) | +0.55 |
| Laws with donor bundle (implementation) | +0.50 |
| Laws with Telegram (implementation) | +0.50 |

Note: the sharpest separation is UN–Research +0.90 against UN–Telegram −0.20 (Williams $p < 0.001$, surviving Bonferroni). Design laws correlate +0.62 with Gov.News, +0.55 with development banks, +0.43 with UN agencies, and +0.34 with research organizations. The two law correlations in the last two rows are statistically indistinguishable at $n = 14$.

The wedge is therefore not a random omission of a few topics. Problems that fit formal monitoring, donor priorities, or national policy categories appear in formal channels. Problems that emerge through local implementation, coercion, contractors, or district-level administration

stay concentrated in Telegram and Kun.uz. Within a problem, the law speaks in the donors' terms. The formal channels are not silent; they are selective.

## 4.3 H3: Whose Information Enters Formal Output?

Even with double vision, formal response may remain adequate if donor expertise and administrative monitoring substitute for farmer feedback. That donors and the state attend to the same problems is not enough to show this. Substitution requires that farmer voice add little to formal response once donor attention is accounted for, and that donors are not simply passing farmer complaints up to the state. In its strongest form, the donor-carried response must do the work farmer feedback would have done. We measure donor attention and farmer voice as each channel's share of problem mentions within a year, and formal response as problem-targeted laws.[18]

### Does the formal problem agenda update during implementation?

The formal agenda changes little with implementation. After rollout, donor documents and laws continue to emphasize much the same problems that dominated during design, and the same pattern holds if implementation is dated from 2023 rather than 2022 (Table 5; the thinner channels are compared in the appendix). What the agenda fails to absorb is the consequential point. During implementation farmers directly encounter contractor fraud, forced adoption, equipment breakdown, and water-timing failures, and see whether the technology pays off, while donors continue to emphasize generic adoption barriers such as cost and training. The channels overlap on high costs and unpaid subsidies; beyond those, contractor fraud is prominent in the chat but absent from donor documents, forced adoption draws substantial farmer but little donor attention, and training shows the reverse (Appendix Table 14; descriptive, since the comparison covers seven locally revealed problems). The formal agenda carried its design-stage understanding into implementation without absorbing much of what farmers revealed.

### Whose signal does lawmaking track?

The evidence points to a donor-anchored but selectively farmer-responsive pattern. The annual panel asks whether a problem receiving more attention in one year receives more problem-targeted laws in the next, comparing the 14 problems within each year. In the main 2021–25 window, prior donor attention is positively associated with lawmaking, while

[18] Inference is permutation- or bootstrap-based alongside cluster-robust asymptotics ($n = 14$ problems; 42–70 problem-year observations depending on the timing window; specifications in Appendix E).

Table 5 Formal problem attention across design and implementation. Cells are Spearman correlations across the 14 canonical problems between each channel's attention profile before and after the cutoff. The first column compares design (2018–21) with implementation (2022–25), with problem-bootstrap 95% intervals in brackets; the second moves 2022 into the earlier period and compares 2018–22 with 2023–25.

| Channel | 2022 implementation boundary | Alternative 2023 cutoff |
|---|---|---|
| Donors | +0.84 [+0.46, +0.98] | +0.86 |
| Laws | +0.89 [+0.58, +1.00] | +0.72 |

Notes: Intervals resample the 14 problems; the alternative cutoff is shown as a point estimate. Counts of texts that mention at least one problem, before/after, are donors 43/57 and 65/35 documents, and laws 38/30 and 48/20 distinct acts, under the two cutoffs. Coverage across the phase boundary is more limited in Gov.News, Telegram and Kun.uz; their profiles and sample counts under both cutoffs are reported in Table 16 (Appendix E).

the farmer-voice association is not detectable under the same permutation test (Table 6). The direct comparison cannot distinguish the two coefficients, and the farmer estimate depends on timing: it appears in some later windows, and at a two-year lag the ordering reverses.[19] The annual evidence therefore identifies a more consistent donor association in the main window, not a statistically larger donor effect.

Table 6 Whose signal does lawmaking track? Problem×year Poisson models of problem-targeted law counts on one-year-lagged attention, with year fixed effects. Cells report coefficients with within-year permutation $p$-values in parentheses.

| Sample | $n$ | Donor attention | Farmer voice |
|---|---|---|---|
| All problems, 2021–25 (main) | 70 | +0.41 ($p = 0.012$) | +0.20 ($p = 0.194$) |
| All problems, 2023–25 | 42 | +0.52 ($p = 0.027$) | +0.40 ($p = 0.083$) |

Notes: Attention variables are standardized within each row's estimation sample; standard errors are clustered by problem. Permutations jointly shuffle donor and Telegram attention across problems within each year. The panel starts in 2021 because the chat opens in late 2020, so lagged voice for 2020 is unobserved; dropping the structurally zero 2021 lagged-voice rows leaves the estimates essentially unchanged. For the main row, cluster-robust $p$-values are 0.015 (donor) and 0.216 (voice); for 2023–25 both are below 0.001, though the within-year permutation values in the cells remain the primary gauge. The exact test of the donor minus voice coefficient difference (+0.21) enumerates all $2^{14}$ per-problem swaps and gives two-sided $p = 0.409$ (cluster-robust Wald $p = 0.287$).

Farmer voice is nevertheless not redundant. Over the implementation phase as a whole, adding farmer attention to a donor-only model improves fit, just as adding donor attention improves a voice-only model, under all three inference procedures.[20] This conditional contribution is the clearest evidence that some farmer information enters formal response, and the problem-

[19] Two-year-lag and lag-averaged estimates are in Appendix E. Horizons beyond two years are right-censored, since voice peaks in 2024–25 and the law series ends in 2025; the phase-level comparison below and the two untouched problems farmers raised most, forced adoption and low yield/profit, need no horizon assumption.

[20] The nested-model $p$-values and AIC values are in Appendix E. We estimate no design-phase model: Telegram's design-phase profile rests on 25 messages (Table 16, notes).

set comparison shows its limits. The law's overall resemblance to Telegram is concentrated in high cost and unpaid subsidies; without them, the law–Telegram association weakens while law–donor alignment remains, and across the 14 problems donor attention is unrelated to farmer attention ($\rho = -0.02$; Table 13, Appendix D). Across the locally revealed set plus weed infestation, and across the seven locally revealed problems themselves, neither law correlation is clearly different from zero (Appendix Table 14). Farmer information therefore appears to enter mainly where it overlaps with problems the formal system already recognizes, rather than redirecting lawmaking toward the failures revealed locally.

Selective incorporation could occur directly or because donors relay farmer complaints. The relay test cannot distinguish these possibilities: donor attention follows lagged farmer attention positively but imprecisely ($b = +0.16$, 95% CI $[-0.09, +0.42]$), and the permutation and wild-cluster procedures disagree ($p = 0.048$ and $p = 0.250$).[21] The evidence thus supports a legislative agenda organized around the donor-aligned schema, with farmer voice entering selectively inside it; it does not show that donors carry farmer information upward or that their effect is statistically larger.

**Whose version of the problem enters the law?**

Agreement on a broad problem label does not necessarily imply agreement about what the problem consists of. We divide each of the fourteen problems into concrete sub-aspects (Appendix E) and classify every selected Telegram message, donor document, and law against the codebook of its problem, adding Gov.News, Kun.uz, and the FGD transcripts as further sources. The primary labels take the union of the Qwen3.8 and Luna classifications of the full texts; the model-specific results provide robustness checks. Two tests follow. Problem by problem, we compare the sub-aspect distributions using base-2 Jensen–Shannon divergence, defining $\Delta$ as the law's divergence from donors minus its divergence from Telegram, so that negative values mean the legal texts are closer to the donor account (bootstrap intervals; specifications in Appendix E). Pooling problems, we ask whether a sub-aspect's share of donor attention or of farmer attention predicts how many laws address it, holding the problem fixed. We begin with the three original shared problems, high cost/loan burden, no training/advice, and water supply shortage.

On *no training/advice*, farmers report unfamiliar features, absent demonstrations, and installers who provide little consultation (operator know-how and supplier service account for 79% of Telegram assignments), whereas donors propose demonstration areas, extension programs, and capacity building, and laws prescribe seminars, qualification courses, and

[21] Cluster-robust $p = 0.20$.

scientific guidance. The legal distribution is much closer to donors ($\Delta = -0.186$, 95% CI $[-0.346, -0.096]$), with the same conclusion in both model-specific checks.

On *high cost/loan burden*, farmers write from their balance sheets (a 650-million-UZS loan, a 150-million-UZS transformer and power line, pumps, reservoirs, electricity), donors frame the constraint through project finance (one report records some 300 subloan applications rejected for insufficient collateral), and laws specify credit terms, collateral, exemptions, subsidies, and compensation. The legal distribution is again closer to donors ($\Delta = -0.097$, 95% CI $[-0.150, -0.051]$) under both robustness checks.

On *water supply shortage*, farmers ask whether water reaches the field at all, describing absent sources, failed connections, leaking canals, upstream withdrawals, and falling groundwater. Donors emphasize deteriorated canals, pump stations, and water-sharing agreements, and laws authorize wells, pumps, pipes, and network reconstruction, so the legal account concentrates on capacity and infrastructure rather than farmers' experience of scarcity and delivery. It is marginally closer to donors ($\Delta = -0.098$, 95% CI $[-0.229, -0.0002]$, an upper bound that crosses zero under a second bootstrap draw), and neither model-specific check separates the two divergences, so we treat the result as suggestive.[22]

The three exploratory problems show the pattern's limits. On *power cost & cutoffs*, farmers describe disconnection, unaffordable bills, and missing transformers, while donors and laws emphasize grid construction and efficient pumps; the law is again closer to donors ($\Delta = -0.239$, Bonferroni-adjusted interval $[-0.351, -0.135]$). On *soil/climate mismatch*, Telegram asks whether the technology fits heat, saline soils, and shallow groundwater, while donor and legal texts speak of assessment, drainage, and crop matching; only five law acts make this comparison descriptive. On *unpaid subsidies*, farmers report long waits and applications moving among offices, whereas laws specify eligibility, documents, and deadlines; only four of 24 donor documents receive a subsidy sub-aspect, leaving that comparison too sparse to separate.

The locally revealed problems are thin on the law side, as is weed infestation: forced adoption, low yield/profit, and weed infestation have no law acts, and the few acts tagged water timing mismatch received no sub-aspect code, so for those four problems the farmer and donor versions can only be described. Letting both slopes of the sub-aspect regression differ between the six shared problems and the locally revealed ones gives the direct test of selectivity announced in Section 3.3: the donor slope is smaller on the locally revealed problems under all three label sets, though the difference is marginal on the primary labels

[22] This comparison includes only texts tagged as water supply shortage through source unavailability. It does not pool the separate water timing mismatch problem; allocation and delivery access is a sub-aspect inside the source-unavailability codebook.

($p = 0.095$), and the farmer slope is not detectably larger there. Among the four that reach law, corruption/fund misuse is the one problem whose legal formulation is closer to the farmer account ($\Delta = +0.12$, 95% CI $[+0.02, +0.29]$): farmers and laws both speak of diverted funds and false installation reports, while donors speak of oversight and vendor capture. Contractor fraud leans toward the donor account, and equipment breakdown and faulty installation lean neither way, but none of the three is separable at eight to eleven acts.

Table 7 pools the problems. The unit is a sub-aspect within a problem, the outcome is the number of law acts carrying it, and the regressors are its shares of donor and of Telegram assignments within that problem; problem fixed effects absorb which problems receive laws at all, so the coefficients compare versions within a problem. Across the ten problems with any law link, a sub-aspect's donor share predicts its law count while its Telegram share does not, and the difference is significant ($p = 0.016$): a ten-point rise in donor share multiplies the expected law count by about one and a half, the same rise in farmer share leaves it unchanged. The association is strongest on the six shared problems. It survives conditioning on the Gov.News share of the same sub-aspect, a domestic institutional text, pooling the FGD transcripts with Telegram, restricting donor documents to those published before the 2022 rollout, and dropping any single problem. Luna-only labels give the same result; Qwen-primary labels, which assign fewer codes, give a weaker donor coefficient whose difference from the farmer coefficient is not significant.

Table 7 Whose version of a problem enters the law? Poisson models of the number of law acts carrying a sub-aspect on the sub-aspect's within-problem share of donor and Telegram assignments, with problem fixed effects. Cells report coefficients with within-problem permutation $p$-values in parentheses; the last column tests the donor minus Telegram difference by per-sub-aspect swaps (5,000 draws).

| Specification | Sub-aspects (problems) | Donor share | Telegram share | Difference $p$ |
|---|---|---|---|---|
| All legislated problems | 68 (10) | +3.82 (0.004) | +0.30 (0.75) | 0.016 |
| Shared six problems | 41 (6) | +6.27 (0.001) | −0.20 (0.83) | 0.001 |
| With Gov.News share | 68 (10) | +3.91 (0.002) | −0.22 (0.82) | 0.007 |
| Farmer share from Telegram and FGDs | 68 (10) | +3.80 (0.004) | +0.37 (0.71) | 0.018 |
| Donor documents before 2022, laws 2022–25 | 68 (10) | +3.10 (0.007) | +0.68 (0.46) | 0.054 |
| Luna-only labels | 68 (10) | +3.74 (0.004) | +0.30 (0.76) | 0.017 |
| Qwen-primary labels | 68 (10) | +2.60 (0.034) | +0.54 (0.65) | 0.13 |

Notes: Union labels except where stated. Problems with no law act carrying a sub-aspect drop out of the fixed-effects model: forced adoption, low yield/profit, and weed infestation have no law acts, and the acts tagged water timing mismatch received no sub-aspect code. The Gov.News share coefficient is +1.16 ($p = 0.11$). Leave-one-problem-out donor coefficients range from +3.0 to +5.5.

This does not imply that farmer voice has no effect on which broad problems receive formal attention. It shows that once a problem enters formal output, its legal formulation follows the mechanisms already legible to the formal and donor systems, and does so most clearly

where donors hold an instrument. Farmer voice may help put the problem on the agenda without determining the version of the problem that the law addresses.

**Does formal response change farmer reports?**

Attention substitution is not the same as functional substitution: does donor attention move laws that, in turn, move farmer reports? Here Telegram is a pure *observer* channel. Donor attention at $t-1$ is positively but imprecisely associated with lawmaking at $t$; lawmaking at $t$ is weakly and imprecisely associated with *more* negative messages at $t+1$ conditional on the current complaint rate; and the reduced form from donor attention to next-period reports is indistinguishable from zero, in the full sample and in a cleaned direct-message sensitivity (coefficients in Appendix E). The chain does not establish mediation, but it agrees with the weekly event study (Section 4.5) on the functional point: neither shows a decline in complaints after formal response, and the two focal unlegislated problems contribute no law events to either test.

Nearly half of the WST laws (157 of 327) carry donor-financing markers, yet the refusal to legislate the politically costly problems is not a donor artifact:[23] forced adoption and the profitability premise receive zero laws in donor-linked and domestic lawmaking alike. Donor substitution, on this evidence, is a claim about legislative attention rather than outcomes: lawmaking follows donor attention, farmer voice adds to it where the two overlap, whether donors relay farmer voice is unresolved, and we detect no subsequent decline in farmer complaints.

## 4.4 Which Problems Receive Weak Formal Uptake?

The direct tests show that legislative updating covaries with donor attention and admits farmer information selectively. We now ask *which* problems travel through formal channels and which do not, using the five-profile policy-learning typology of Section 3. Because law and state-response signals enter the clustering inputs, the typology summarizes selective response rather than testing it, and with 14 problems it is descriptive; the profile names are our interpretive labels. Its sensitivity to the algorithm, the number of clusters, and perturbation is reported in Appendix B, which is why we read the donor-mediated, decay/coercion, and governance-failures boundaries with caution.

Figure 5 shows why H3 is only partly an "absence of response" story. Two profiles are functioning policy-learning loops. In the *donor-mediated loop* (high cost/loan burden, no

[23] An act is donor-linked if its Russian or Uzbek text names a donor or development bank or carries a loan, grant, or international-financing marker (bilingual twins pooled). The rule is an upper bound, since generic financing boilerplate triggers it; donor names alone mark 50 acts (15.3%).

| Profile | Problem bundle | Signal | Design | | Impl. | Pairwise ratings Obs | Solv | Sens |
|---|---|---|---|---|---|---|---|---|
| **WORKING LOOPS** | | | | | | | | |
| donor-mediated loop | High cost/loan burden<br>No training/advice<br>Water supply shortage<br>Soil/climate mismatch | Public voice | 75 | → | 48 | 39 | 50 | 39 |
| | | Donors | 89 | → | 89 | | | |
| | | State | 76 | → | 67 | | | |
| state learning loop | Power cost & cutoffs<br>Unpaid subsidies | Public voice | 43 | → | 71 | 18 | 39 | 64 |
| | | Donors | 54 | → | 66 | | | |
| | | State | 80 | → | 73 | | | |
| **WEAK OR BROKEN LOOPS** | | | | | | | | |
| governance failures | Faulty installation<br>Corruption/fund misuse<br>Contractor fraud | Public voice | 47 | → | 78 | 57 | 57 | 79 |
| | | Donors | 36 | → | 32 | | | |
| | | State | 50 | → | 65 | | | |
| decay / coercion | Equipment breakdown<br>Forced adoption | Public voice | 79 | → | 62 | 79 | 50 | 71 |
| | | Donors | 46 | → | 41 | | | |
| | | State | 32 | → | 34 | | | |
| abandoned / dormant | Low yield/profit *<br>Water timing mismatch<br>Weed infestation | Public voice | 26 | → | 16 | 79 | 43 | 29 |
| | | Donors | 11 | → | 32 | | | |
| | | State | 22 | → | 25 | | | |

Cells: within-channel percentile rank 0–100 (profile median).
Obs / Solv / Sens: higher = harder to observe, harder to solve, more donor-sensitive.

* Low yield/profit is a boundary case: donors recognize it, but the state response remains weak.

Figure 5 Cosine $k = 5$ policy-learning typology. Each row is a policy-learning profile of WST problems. Signal values are profile median within-channel percentile ranks, 0–100, for public voice, donor attention, and state response, from design to implementation. Right-side rating cells report cluster medians of BT-refit pairwise-rating percentile ranks for observability, solvability, and donor-pressure sensitivity; higher values denote harder observation, harder solution, or greater donor sensitivity.

training/advice, water supply shortage, soil/climate mismatch), donor substitution coincides with problems that are donor-legible, donor-financed, and easy to translate into formal response; no training/advice is its ambiguous member, since the sub-aspect test above shows the laws answer the donor version (extension programs) rather than the know-how and after-sales failures farmers report. In the *state-learning loop* (power cost & cutoffs, unpaid subsidies), public voice rises during implementation and state response remains high, so the domestic loop still works. The two succeed through different routes: the unpaid-subsidies problem is the clearest user-initiated loop, with farmer attention rising sharply during implementation, donor attention alongside it, and the second-largest share of problem-targeted lawmaking in both phases; power cost & cutoffs was flagged heavily by donors already at design, before farmers raised it, and its legislative share rose when the problem surfaced on the ground.

The remaining profiles locate where the loop weakens. The *governance failures* profile (faulty installation, corruption/fund misuse, contractor fraud) shows partial recognition without adequate accountability: public voice rises, donor response is weak, and state response is only partial. The *decay/coercion* profile (equipment breakdown, forced adoption) is the state-blind edge of the typology, with *forced adoption* as its core (zero laws in both phases and almost no Gov.News coverage). The *abandoned/dormant* profile is mostly low-signal/low-response, but *low yield/profit* is a boundary case: heavy farmer attention, more donor than state recognition, and no law, plausibly because the claim that drip irrigation does not pay off sits uneasily with the program's win-win narrative. The typology therefore separates donor substitution for generic adoption barriers, domestic policy learning for problems tied to national programs or utilities, and selective breakdown for local, politically costly, hard-to-verify, or narratively threatening implementation failures.

As an independent check, we compare the typology with the pairwise ratings of each problem's observability and solvability. These ratings were elicited separately from the clustering inputs, though by an author-rater who had seen earlier coverage results (Section 3), so agreement between the typology and the rating surface is informative rather than mechanical. Observability separates the loop classes more clearly than solvability does: working-loop problems are more visible to the formal system, while failed- and partial-loop problems concentrate in less observable regions (Figure 6).

Low observability, however, has two sources. The first is political cost. Corruption/fund misuse, contractor fraud, forced adoption, and faulty installation are donor-sensitive and entangled with accountability chains. They are hard to solve because addressing them imposes political costs, and hard to surface because the actors who would report them may have reason not to. The second is local fragmentation. Weed infestation, water timing

mismatch, and equipment breakdown are technically more tractable, but they spread across farm-level events that no centralized channel aggregates; here the constraint is monitoring infrastructure rather than active suppression. Low yield/profit is again the exception: relatively observable, but weakly answered because it has no simple technical answer and contests the program's premise. Appendix Figures 10 and 11 trace the same problems dynamically, showing lawmaking contracting on the observable dockets while public voice surged on the governance failures.

Removing the law signals leaves the partition unchanged, although removing Telegram or donors changes it materially (Appendix B).

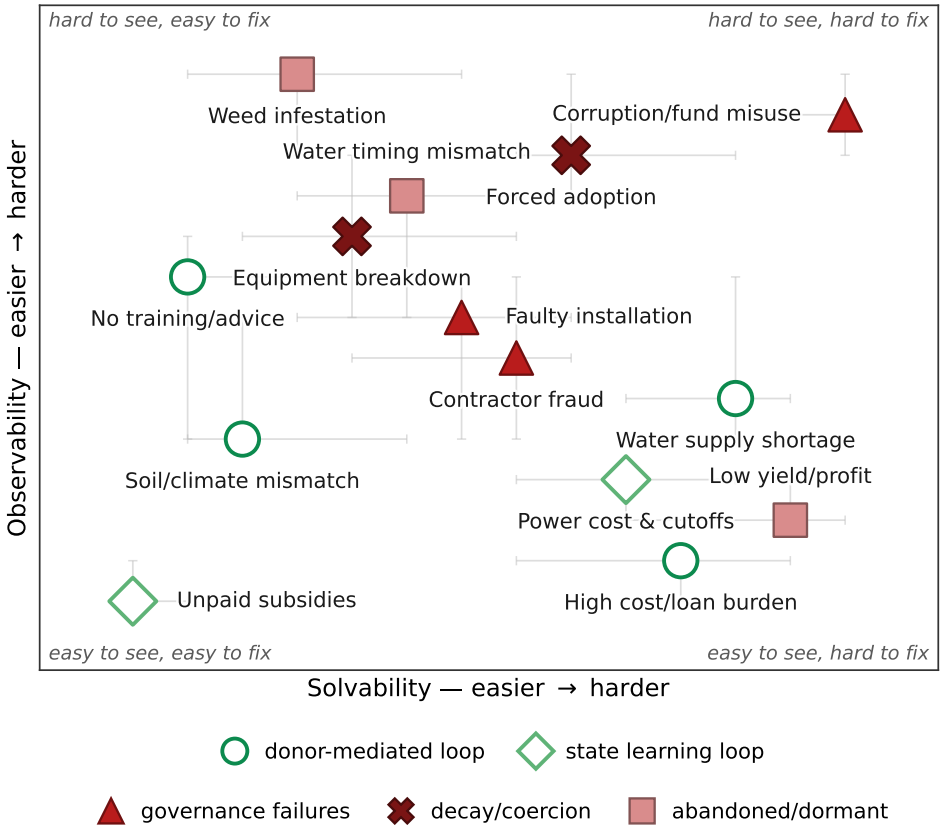



Figure 6 The 14 WST problems on the Observability × Solvability surface, colored by the five-profile typology: green shades mark the two sustained-loop profiles, red shades the three broken-loop profiles. Sustained profiles are drawn open (white-filled) and broken profiles solid (the same open/solid loop-state encoding as Figures 10 and 11) while marker shapes separate profiles within each family. Positions are from the BT-refit pairwise ratings, with both axes running from easier to harder and 80% Laplace CI bars; overlapping bars mean adjacent ranks are not separable: the grouping, not the fine ordering, is the signal. The ratings were elicited separately from the clustering inputs (see Section 3 for the elicitation caveats), so coherence between color and position is an informative check.

## 4.5 Robustness, Blame, and Placebo Evidence

Three checks bear most directly on the qualitative reading. First, blame attribution asks whom farmers hold responsible. We classify 1,442 implementation-phase Telegram message-by-problem pairs (13 non-grievance pairs excluded; Table 8, problem-level detail in Appendix A). The governance-failures profile is primarily an accountability-chain problem between farmers and private actors: contractor fraud draws overwhelmingly private blame, while corruption/fund misuse bridges state and private attribution. Forced adoption draws mostly state blame, yet the formal response remains weak, and the state-learning loop carries the highest state-blame share while the state continues to respond, so state blame by itself does not predict policy failure. Failure appears where blame is tied to local coercion or to state–firm accountability chains that formal channels only partly absorb. Negativity alone does not predict it either: 81.2% of problem-bearing implementation-phase Telegram messages are negative against 17.8% of problem-bearing Gov.News articles (text-level shares; Table 8 reports pair-level shares), and profile-level negativity is uniformly high, so the wedge of H1 is a difference in problem mix, not merely in tone.

Table 8 Sentiment and blame attribution by policy-learning profile, 2022–2025. Pairs are classified text–problem pairs; Base is the blame denominator, which excludes non-grievance pairs and retains systemic and no-blame cases. NEG% is the share of pairs whose text is classified negative.

| Policy-learning profile | Pairs | Base | NEG% | State% | Private% | Systemic% | No blame% |
|---|---|---|---|---|---|---|---|
| Donor-mediated loop | 358 | 353 | 80.2 | 26.6 | 20.1 | 41.6 | 11.6 |
| State-learning loop | 447 | 445 | 86.1 | 61.8 | 4.5 | 32.8 | 0.9 |
| Governance failures | 313 | 311 | 92.7 | 18.0 | 70.4 | 10.3 | 1.3 |
| Decay/coercion | 205 | 203 | 80.5 | 53.2 | 14.8 | 16.7 | 15.3 |
| Abandoned/dormant | 119 | 117 | 72.3 | 25.6 | 12.8 | 23.1 | 38.5 |
| Gov.News baseline | 67 | 59 | 26.9 | 33.9 | 33.9 | 30.5 | 1.7 |

Second, a weekly event study around exact enactment dates covers the 63 implementation-phase act×problem links (23 acts) with complete ±26-week Telegram windows. Complaints dip in the announcement window and rise nominally after enactment in the main specification, but the rise weakens once decree-discussion threads and news accounts are removed, disappears in a paired nonparametric check, and does not cross the empirically calibrated small-cluster threshold. No full-sample specification provides evidence of a post-law decline, but precision is weak: the primary unique problem–calendar-week simulation has 80% power only for a sustained decline of about 80.0%. Estimates, the importance split, and the seasonal positive control are in Appendix A.

Third, the topic placebo tests the main identification concern directly. If Telegram were simply always more grievance-heavy than official media, or if Gov.News were always less specific, the same wedge should appear in more neutral policy topics. Figure 7 shows that it does not: WST has a sustained implementation-phase rise in Telegram problem attention alongside a sharp decline in official problem attention, while climate, pests, and Russia–Ukraine war consequences do not reproduce the same divergence. Together with the Kun.uz bridge, sentiment results, and blame classifications, the placebo supports interpreting the WST pattern as a policy-specific implementation signal rather than a generic media asymmetry. Appendix Figure 12 plots the Telegram−Gov.News gaps on a single shared axis, where the contrast is sharpest: the WST gap widens through implementation while adverse weather reverses sign outright.

# 5 Discussion: Revisiting the Kaleidoscope Model

The results revise, rather than overturn, the KM interpretation of the WST subsidy: the citizen–state feedback loop is partially, but not fully, broken. At design, donor attention is visible while the farmer signal is barely observed, so substitution there describes the information that was available rather than a contest that voice lost. We estimate no design-phase farmer-voice model. After rollout, farmers held more of the relevant information and the phase comparison favors a combined donor-and-voice model, yet legislative updating remained associated with donor attention, and within problems the law's formulation follows the donor version, most clearly for training, financing, and electricity, with no detectable donor tilt on the locally revealed problems that reach law at all. Formal output is followed by no detectable reduction in farmer complaints. Policy response is therefore selectively misaligned for the implementation failures least likely to move through formal channels, and even where attention is aligned no large improvement is visible on the ground (smaller ones would escape our gauge; Section 4.5).

Table 9 re-evaluates the KM determinants stage by stage, setting the formal-source benchmark of Section 2 beside the assessment once view-from-below information is admitted. Agenda-setting and adoption largely survive: the relevant problem was recognized, the focusing event and propitious timing held, and government veto players did not block the reform. The largest revisions concentrate in design, implementation, and evaluation, and the downgrades largely track the profiles where formal channels fail to absorb farmer signal.

The information wedge also changes how ongoing policy shifts should be read. Since 2024, Uzbekistan has sharply scaled up budgetary investment in lining and reconstructing

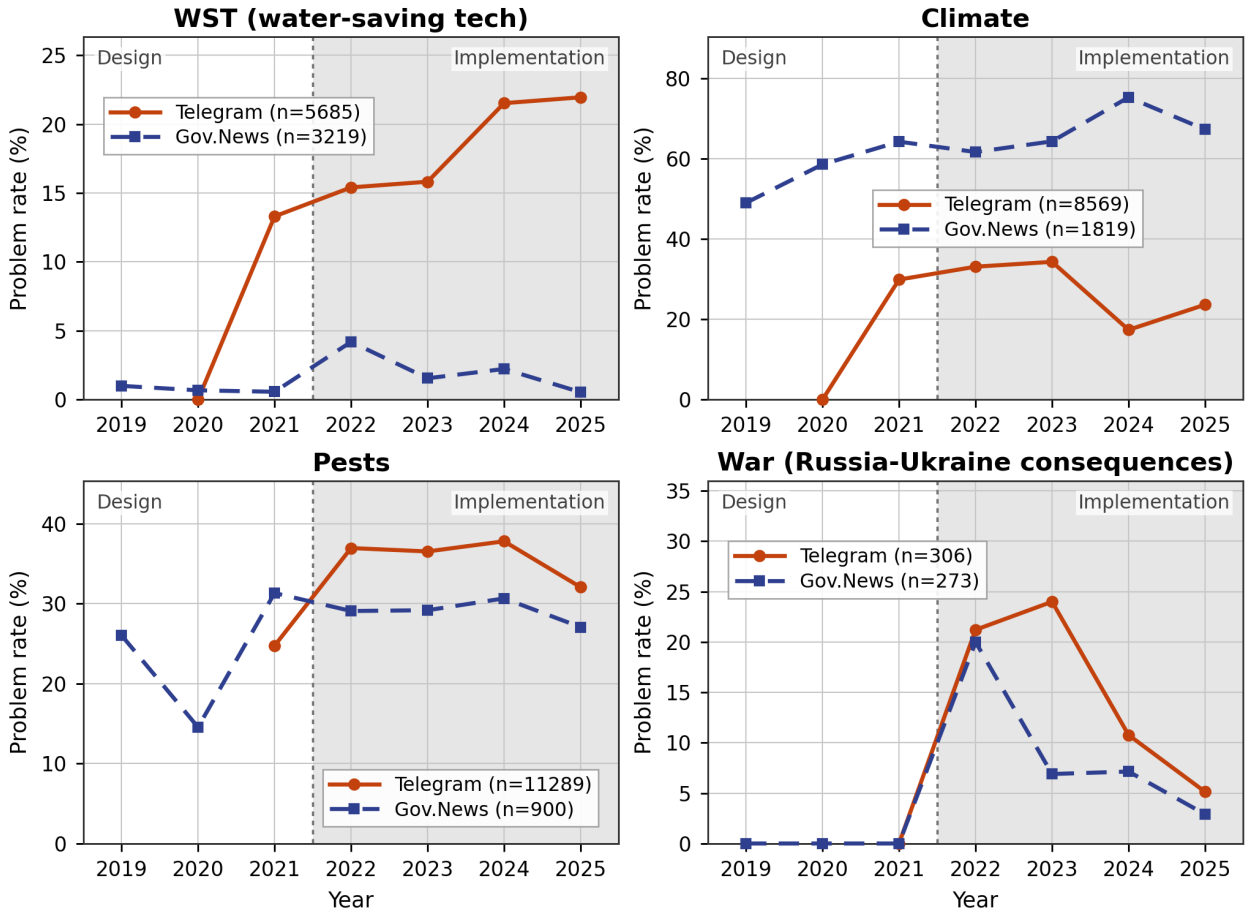



Figure 7 Topic placebo for the Telegram–Gov.News divergence. Lines show the share of topic-positive texts with at least one problem mention, by source and year, for WST and three placebo topics: adverse weather/climate; pests and crop disease; and consequences of the Russia–Ukraine war. The dashed line at 2021.5 divides the design from the implementation phase, and the implementation window is shaded. WST shows a sustained implementation-phase rise in Telegram problem attention while official problem attention declines; the more neutral placebo topics do not reproduce the same implementation wedge. Panel $y$-scales are independent, and topic schemas differ, so levels are comparable within panels only.

Table 9 KM determinants of policy change: the formal-source benchmark and the revised assessment with view-from-below information.

| **Policy stage** | **Determinant** | **Benchmark** | **Revised** | **Evidence producing the revision** |
|---|---|---|---|---|
| Agenda setting | Recognized relevant problem | + | + | — |
| | Focusing event | + | + | — |
| | Powerful advocacy coalitions | + | +/− | Farmer-side coercion enters the record |
| Design | Knowledge & research | + | +/− | Contractor quality, coercion, and realized profitability unobserved from above |
| | Norms, biases, ideology and beliefs | − | +/− | Elite win-win norms persist from above; farmer profitability and cost doubts sharpen from below |
| | Cost-benefit calculation | + | +/− | Uncompensated power costs; contested profitability premise |
| Adoption | Powerful proponents/opponents | + | +/− | MoWR canal preference; upstream farmers (Section 2, Appendix A) |
| | Government veto players | + | + | — |
| | Propitious timing | + | + | — |
| Implementation | Requisite budget | + | +/− | Unpaid subsidies; loan-interest grievances |
| | Institutional capacity | + | +/− | Governance-failure profile (fraud, faulty installation) |
| | Implementing stage veto players | + | − | Local coercion; contractor selection |
| | Commitment of policy champions | + | +/− | Coercion and payout delays enter the record |
| Evaluation & Reform | Changing information and beliefs | + | +/− | Donor-legible and locally implemented problems diverge |
| | Changing material conditions | + | +/− | Profitability premise contested from below |
| | Institutional shifts | 0 | 0 | — |

Note: + indicates the determinant favored successful policy change in this episode (for veto players, that none blocked the reform); − that it worked against success; 0 neutral or no change; +/− mixed evidence. The benchmark column codes the episode from formal sources alone (Section 2); the revised column admits the view-from-below channels.

irrigation canals.[24] Without the view from below, the natural reading is that the WST program is functioning well and the government can turn to other aspects of water efficiency. With it, the same shift is also consistent with the subsidy's lack of robust success allowing those opposed to drip irrigation, for strategic reasons, to push their own agenda; the recent acts scale canals and WST together, so the margin of interest is the redistribution of budget across the competing ministries' preferred instruments rather than any abandonment of WST. Likewise, an external observer would read the post-2022 decline in problem-targeted lawmaking and official coverage as success; with the view from below, the same decline is consistent with a feedback loop that stops updating precisely where the hardest implementation problems concentrate.

Why does the informational advantage shift after rollout, and why do donor-legible problems fare better? The design-phase strength of donor substitution is also its limitation. Donors carry generic, policy-playbook information that converts easily into instruments the formal system already owns, and they have weaker incentives and instruments to convert locally generated failures, such as coercion, contractor fraud, or the claim that drip irrigation does not pay off, into formal response; donor documents flag that last problem more readily than the state does, but it never travels into law. Whether donors relay farmer voice is unresolved, and on the problems implementation revealed their attention runs opposite to farmers' in point estimate, so a state that listens only to donors has no assurance of hearing the view from below. Breakdowns concentrate where the signal is hard to verify centrally or politically costly to act on, so the binding constraint is the channel that carries local information into formal response, not farmer voice as such.

A durable fix would require mechanisms that move credible, verifiable information from farmers to the ministries and protect those who report coercion or fraud. Because the infrastructure and credit constraints documented above interact with the limited installer competition farmers describe, coordinated responses are more likely to work than single-instrument fixes. One tentative institutional implication follows from the misaligned incentives of Section 2: tying both ministries' success metrics and budgets to overall water efficiency, rather than to specific technologies, could help realign the loop, though we do not evaluate such a reform here. The design also yields a falsifiable forward prediction. Because slow absorption of farmer signal beyond a two-year horizon is right-censored in our sample, a feedback loop that is merely slow rather than selective would require lawmaking in 2026–27

[24] PP-454 (24 December 2024) directs a 1.7-fold increase in funds for repairing and concreting irrigation networks; PP-250 (15 August 2025) programs 2,551 km of reconstruction over 2025–2028; PP-79 (5 March 2026) earmarks 480 of 600 billion UZS for canal concreting. The World Bank approved $200 million for lining 259 km of primary canals in May 2025.

to pivot toward the 2023–25 farmer agenda of coercion, contractor accountability, and the profitability premise; the continuously updated law corpus makes this directly checkable.

**Limitations.** Several limitations bound these conclusions. The problem-level analysis rests on only $n = 14$ problems, so the cross-channel correlations and the typology are descriptive and low-powered, and several source contrasts cannot be distinguished at this resolution (Appendix D). Kun.uz contributes only four design articles and 12 implementation articles that mention at least one of the 14 problems, so its problem-mix ranks rest on a small number of labels. The observability and solvability ratings are not a validated measurement. The post-law event windows are correlational (consistent with misalignment but not a causal estimate of legislative effects). The complaint stream is a blunt outcome gauge, so the absence of a post-law decline rules out large improvements more clearly than modest ones.

Our sources are not measured on a common denominator, which is why our inferences rest on rankings and schema alignment rather than on prevalence levels. The farmer-facing channels are self-selected rather than representative, and participants in either Telegram or the FGDs/IDIs may misreport. The event-study appendix separately removes the dozen high-volume accounts that supply about a third of decree-related content; the estimated post-law increase weakens and remains specification-sensitive rather than becoming evidence of a decline (Appendices A and E). The overlap between the focus groups and the chat, together with the consistency of the differences we observe across independent diagnostics, gives us confidence that there is more signal than noise in the data.

# 6 Conclusion

Uzbekistan's water-saving technology subsidy is a favorable case for policy learning: a reform-oriented government, a win-win technology, and an engaged donor community. Read from formal sources, the Kaleidoscope Model scores the program a success. Reading five information channels and the legislative output against one fixed problem schema changes that assessment. Donor knowledge can help fill an information vacuum during policy design, but we find no evidence that it replaces the local implementation information and accountability carried by policy users.

Two findings carry the argument. First, farmer-facing and state–donor sources organize implementation problems into systematically different schemas, the gap widens after rollout (descriptively; Appendix D), and below the problem level the farmer accounts describe each problem in their own terms while the law's terms mirror the donors'. Second, the formal legal

response is selective: legislative attention covaries with donor attention and within legislated problems the law's formulation more closely follows the donor's account.

For development policy, the implication is that donor knowledge should be treated as design-stage knowledge. Donors can anticipate generic adoption barriers, and formal systems convert donor-legible problems into instruments readily. After rollout, the information that matters is held by policy users, and we find no robust evidence that donor attention carries it upward: the permutation and wild-cluster relay tests disagree. Governments that want policies to survive implementation need channels that move credible local information to the ministries and protect those who report coercion or fraud; donors should be cognizant that development finance may orient formal attention toward donor-legible problems.

Are these lessons generalizable? Little in the mechanism is specific to Uzbekistan. It requires accountability mechanisms that remain imperfect even under reform-minded leadership, a donor community that readily accepts evidence shaped like the global template, and a state with some control over how the policy is perceived. Where those conditions hold, favoring perception over reality is cheap, and rent-seeking opportunities can follow. We would therefore expect similar patterns in other donor-supported reforms where local information is constrained.

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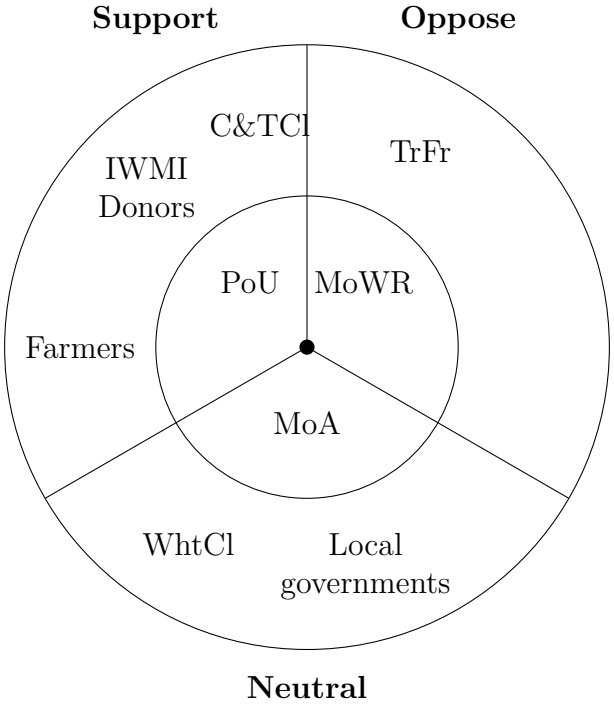
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# A Additional Figures and Tables

This appendix collects supporting displays: the policy chronology and circle of influence, the bilingual law-corpus reliability details, the full cross-channel correlation matrices, the attention-trajectory figures, the blame-attribution problem-level discussion, the detailed event-study results, and the topic-placebo gap series.

## Policy chronology and circle of influence

Figure 8 Circle of influence for the WST subsidy program. The inner circle holds actors with direct decision-making authority; the outer circle, actors with indirect influence. Sectors group actors by stance toward the program.




**Actors in the circle of influence.**

- **Supporters:** IWMI, a research-for-development organization promoting sustainable water management, and the donor community (World Bank, ADB, GIZ) fund and promote water-use efficiency. Progressive farmers adopt modern irrigation to raise productivity, and the PoU sets ambitious adoption goals from the top. Cotton & textile clusters (C&TCl) are supportive in principle, since WST can raise yields and meet sustainability standards, but rapid adoption timelines and uncertain long-term benefits place them close to opposition.
- **Neutral:** The MoA balances traditional practices against new technologies. Local governments are the key implementers but lack a unified stance and are overburdened

Table 10 Drip irrigation subsidy policy chronology. Authors' compilation from the acts cited in Sections 2 and 5; structure adapted from Resnick et al. (2018).

| Year | Policy Events | Political Events | Economic Events | Other events |
|---|---|---|---|---|
| Pre-2016 | | | | International recognition of water inefficiency |
| 2016 | | Death of Islam Karimov, Uzbekistan's first president; Shavkat Mirziyoyev elected as the new President | | |
| 2017 | Broad institutional reforms initiated | | | IWMI research field trial in the Karshi Steppe |
| 2018 | Ministry of Water Resources established; Government endorses subsidies for WST (PP-4087) | | | |
| 2019 | Subsidy program introduced (Cabinet Res. 103 and 664; PP-4499) | Strategy to transition to Green Economy | | IWMI recommends shift away from energy subsidies to WST |
| 2020 | Water Sector Development Concept for 2020–2030 Decree No. UP-6024; PP-4919 launches the drip-irrigation subsidy at scale (December) | | | |
| 2021 | The Strategy for Managing Water Resources and Developing the Irrigation Sector for 2021–2023 was approved (PP-5005, February) | Mirziyoyev reelected (October); enhanced Paris Agreement pledge submitted at COP26 | | |
| 2022 | PP-144 adjusts the subsidy rollout | | | |
| 2023 | UP-27 directs saving of 7 billion $m^3$ of water and further WST expansion | Mirziyoyev reelected in snap election following constitutional referendum | | Official reports indicate significant progress |
| 2024 | Canal-concreting drive launched (presidential instruction, November 2023); PP-454 (December) raises irrigation-network concreting funds 1.7-fold for 2025–27 | | | |
| 2025 | PP-250 (August) programs 2,551 km of irrigation-network reconstruction for 2025–28 | | | World Bank approves $200 million for lining 259 km of primary canals |

by the pace mandated from above, which places them near the neutral–opposed boundary. Wheat clusters (WhtCl) see limited direct benefit from the current focus on drip irrigation and laser leveling.

- **Opposers:** The MoWR, responsible for water distribution and infrastructure, favors canal systems over farmer-level WST (see the water-energy nexus in Section 2). Traditional upstream farmers (TrFr) still perform well with abundant water and flood irrigation and resist adoption.

## Law corpus: bilingual records and reliability

Of the 462 bilingual law records, 135 acts appear twice, once in Russian and once in Uzbek, and are collapsed into single acts by act type, document number, and enactment date. Where the classifier labeled the two language versions of the same act differently (41 of 135 twin pairs), we take the union of the assigned problems. Because the underlying act is identical, twin disagreement doubles as a free test–retest reliability estimate for the classifier on legal text: label sets agree exactly for 94 of 135 pairs (70%; mean pairwise Jaccard 0.75), with per-problem positive agreement reported in Table 11. The estimate is conservative, since it bundles classifier noise with genuine RU/UZ translation differences, and the union basis means a problem detected in either language version enters the act-level counts. Twelve further acts (18 records) dated 2026 fall outside the analysis window. All law counts in the paper are on the 327-act distinct-act basis. Per-problem law counts on the record and act bases are nevertheless nearly perfectly rank-correlated (Spearman $\rho = 0.995$; `appendix_checks/src/law_basis_rank_rho_canonical.py`), and the five-profile partition is unchanged (adjusted Rand index 1.0). Three problems (forced adoption, low yield/profit, and weed infestation) have no tagged twin pairs, consistent with their zero or near-zero law counts, so the twin design provides no reliability evidence for precisely those tags, and it measures stability rather than accuracy in any case.

## Cross-channel correlation matrices

## Attention trajectories and dynamic surfaces

Donor and view-from-below attention belong to two distinct schemas (Table 4), so a problem's standing on one axis says little about its standing on the other, and all four quadrants of their joint plane are occupied. Figure 10 maps each problem's design-to-implementation move in this plane, with bubble size marking its total problem-targeted lawmaking.

Figure 11 reads the same surface dynamically: holding fixed each problem's position on the observability and solvability axes, it plots only what changed from the design to the implementation phase, the formal response (top) and the two within-source attention percentile ranks (bottom). Arrow length encodes each percentage change against a shared 25/50/100% ruler (raw decree links in the top panel, within-source percentile ranks in the bottom), and the two arrows beyond ±100% carry a scale-break tick and print their exact value; labels in the top panel carry raw decree counts, so large percentage swings on small dockets (e.g. contractor fraud, $2 \to 6$) read honestly next to the shrinking large dockets. The

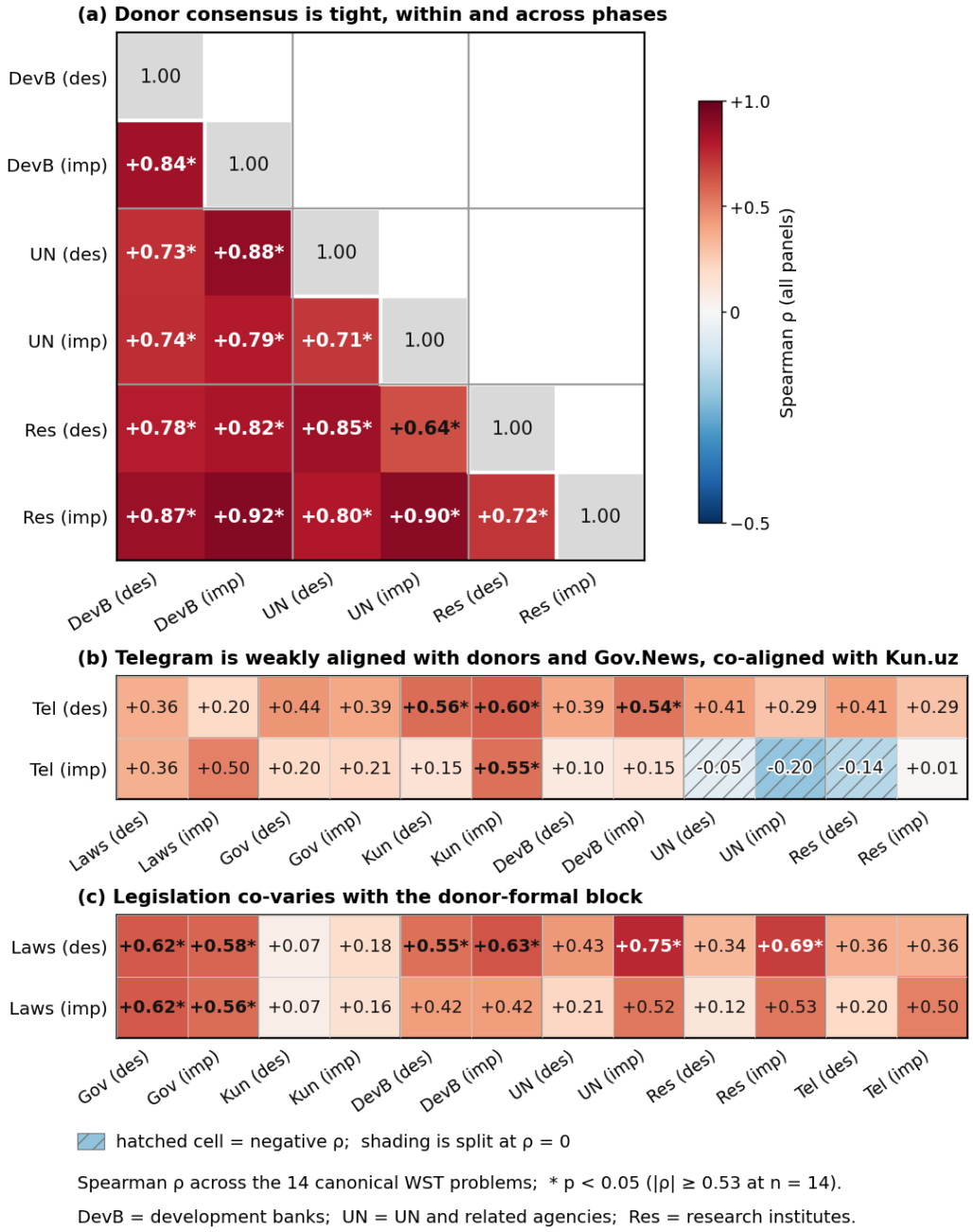



Figure 9 Cross-channel problem-schema alignment, design and implementation phases. The lower triangle reports Spearman correlations across the 14 canonical WST problems ($*p < 0.05$; $|\rho| \geq 0.53$ required at $n = 14$; displays are rounded to two decimals and stars follow the exact $p$, so a printed 0.53 can fall just short, as for Laws–Res in implementation, $\rho = 0.531$, $p = 0.051$). Laws enter as distinct act×problem links; years before a channel begins are treated as missing, not zero. Donor sub-channels are DevB (World Bank, ADB, AIIB, EBRD), UN (FAO, UNDP, IFAD, GEF), and Res (IWMI, ICARDA, GGGI, SDC); these analytical sub-channels regroup the organizations listed in Section 3.1 (IFAD and GEF with the UN agencies; SDC and GGGI with the research organizations).

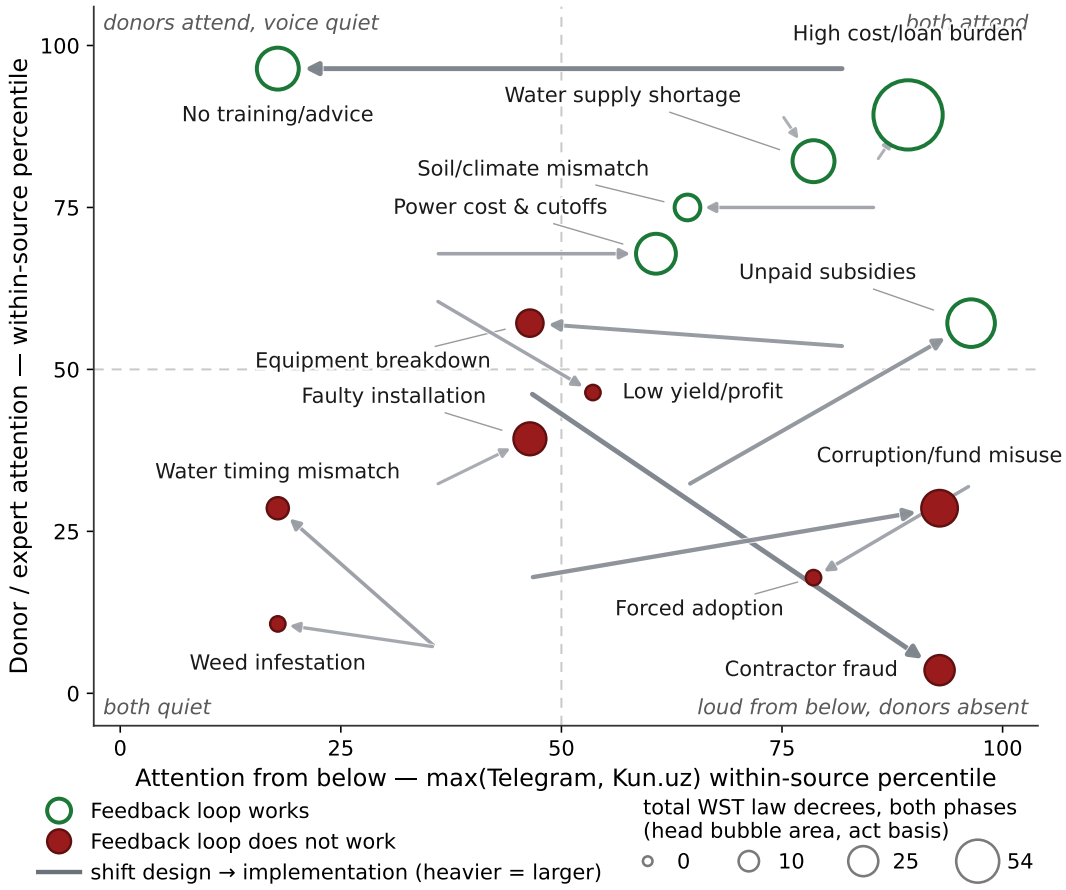



Figure 10 Attention trajectories and legislative intensity. Each problem moves from its design-phase (2018–21) to its implementation-phase (2022–25) position in the attention plane: attention from below ($x$, the larger of the Telegram and Kun.uz within-source percentiles) against donor/expert attention ($y$). Bubble area is the problem's total WST decree count across both phases (act basis, $94 + 81 = 175$ problem–decree links); fill marks the binary loop state (open green circles sustained, solid dark red broken) with profiles as classified in Figure 5. Lawmaking concentrates where donor attention is high. Percentiles are computed within each (source, phase); coincident trajectory origins are dodged slightly for visibility. $n = 14$, descriptive.

Table 11 Test–retest reliability of the law classifier from RU/UZ bilingual twins: 135 pairs in which the same decree was scraped as separate Russian and Uzbek documents and classified independently. Label sets agree exactly for 94/135 pairs (70%; mean pairwise Jaccard 0.75); positive agreement = pairs where both twins carry the problem over pairs where at least one does. The estimate is conservative (it bundles classifier noise with genuine translation differences); the act basis used throughout takes the union of twin labels.

| Problem | Both | One-sided | Tagged pairs | Pos. agreement |
|---|---|---|---|---|
| High cost/loan burden | 17 | 26 | 43 | 0.40 |
| No training/advice | 10 | 4 | 14 | 0.71 |
| Unpaid subsidies | 5 | 10 | 15 | 0.33 |
| Water supply shortage | 6 | 10 | 16 | 0.38 |
| Power cost & cutoffs | 6 | 6 | 12 | 0.50 |
| Corruption/fund misuse | 2 | 8 | 10 | 0.20 |
| Faulty installation | 1 | 7 | 8 | 0.13 |
| Equipment breakdown | 2 | 5 | 7 | 0.29 |
| Contractor fraud | 2 | 3 | 5 | 0.40 |
| Soil/climate mismatch | 2 | 2 | 4 | 0.50 |
| Water timing mismatch | 1 | 2 | 3 | 0.33 |
| Forced adoption | 0 | 0 | 0 | – |
| Low yield/profit | 0 | 0 | 0 | – |
| Weed infestation | 0 | 0 | 0 | – |

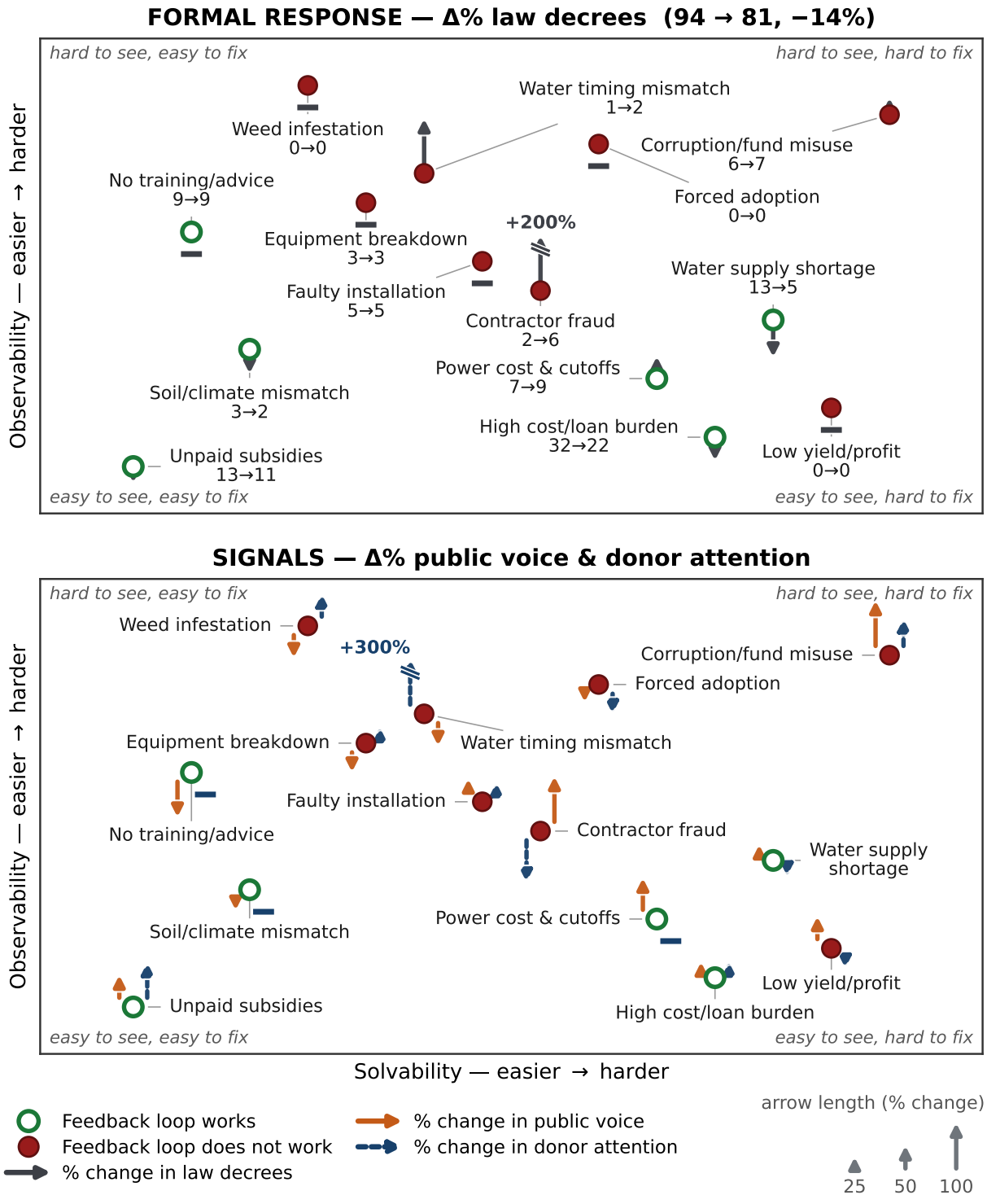
selective pattern reappears as movement: lawmaking contracted on the large, observable dockets (high cost/loan burden 32 → 22, water supply shortage 13 → 5) while public-voice rank doubled for both contractor fraud and corruption/fund misuse and fell for no training/advice (−78%) and equipment breakdown (−43%); donor-attention rank moved opposite to voice on 6 of 14 problems, most sharply on contractor fraud (−92%).

## Blame attribution: problem-level detail

**Classifier.** To distinguish whom farmers hold responsible, we classify blame in implementation-phase text–problem pairs into state actors versus private actors, alongside systemic, no-blame, and non-grievance categories. The classifier is Qwen3.8-27B (a separate model from the ensemble's Qwen3.5 member), run locally with deterministic decoding (temperature 0, fixed seeds). Sentiment is classified by text and blame by text–problem pair. Long texts are split into 4,000-character windows with 500-character overlap, aggregated by plurality for sentiment (ties resolve to Neutral) and by a confidence-weighted vote for blame. We report the 2022–2025 Telegram and Gov.News texts.

To separate accountability mechanisms, we classify 1,442 implementation-phase Telegram message-by-problem pairs and 67 Gov.News article-by-problem pairs. Thirteen Telegram and eight Gov.News pairs are non-grievance, leaving blame bases of 1,429 and 59, respectively; systemic and no-blame pairs remain in those bases.

The sentiment column shows that Telegram problem reports are predominantly negative across all profiles, so negativity alone does not predict policy failure; the problems muted in




Arrow up = grew after rollout, down = shrank; short dash next to a dot (in its series colour) = no change (incl. 0→0); thin grey lines connect displaced labels to their dots. Arrow length = % change on the shared ruler at right (same scale in both panels); arrows with a break tick exceed ±100% and print their exact value; changes under ~15% are drawn at a minimum visible length.

Figure 11 What changed after rollout. *Top:* problem–decree links; *Bottom:* within-source attention percentile ranks (donor arrows dashed), both as design→implementation percentage change. Dots: open green = sustained, solid red = broken loop.

the view from above are not differentially negative, which supports reading the H1 wedge as a difference in problem mix rather than tone. The governance-failures profile is primarily an accountability-chain problem: contractor fraud has 91.9% private blame (base $n = 161$), faulty installation 77.4% private blame (base $n = 53$), and corruption/fund misuse bridges state and private attribution (43.3% state, 30.9% private; base $n = 97$). The decay/coercion profile is different: forced adoption draws 66.7% state blame (base $n = 147$), yet the formal response remains weak. This should not be read as a purely district-level mechanism or as a claim that only state actors matter. In farmer accounts, local state authority and installer markets often appear together: district-level actors can shape access to firms, while selected firms carry out installation and service. We therefore read the state-blind edge as a local implementation-network problem, where state authority, firm selection, and accountability are difficult to separate.

## Event study: detailed results

A weekly event study around exact enactment dates uses all 63 implementation-phase act×problem links (23 acts) with complete ±26-week Telegram windows. Bilingual records are collapsed by unioning their problem labels, and decree-referencing posts, reply threads, and high-volume news accounts can be removed from both numerator and exposure. In the primary stacked Poisson model, the announce-window rate ratio is 0.843 ($p = 0.033$) and the post-window ratio is 1.275 ($p = 0.009$). Removing direct law references gives 0.869 ($p = 0.094$) and 1.296 ($p = 0.010$), respectively. Removing threads gives a post ratio of 1.293 ($p = 0.029$); removing news accounts gives 1.212 ($p = 0.088$); removing both gives 1.209 ($p = 0.120$). A paired nonparametric check is flatter (median post/pre ratio 1.018, $p = 0.739$, 49 usable events). Thus no full-sample specification provides evidence of a complaint reduction; evidence for an increase is specification-sensitive.

Two diagnostics bound the gauge. First, the pre-specified positive-control criterion is a joint verdict over the season-bound problem group: Fisher-combined seasonality significance (met, $p = 0.002$), correct April–September phasing for every eligible season-bound problem (not met), and a median seasonal amplitude exceeding the administrative group's (met); the phasing clause is the one that fails. Electricity complaints themselves retain the predicted July peak and zero January counts, with a within-year permutation $p = 0.024$, short of the 0.01 mark recorded in the archived claim check. Second, the importance split contains 15 major and 8 minor acts (37 and 26 events). Major laws have a deeper announce-window dip (interaction $p = 0.017$), but the post-window interaction is not significant ($p = 0.152$). The major stratum rises after enactment (rate ratio 1.68, $p = 0.020$) while the minor stratum falls (0.77, $p = 0.036$), but the interaction is not significant and both strata are small. The primary unique-cell NB2 simulation shows why the nominal event-study $p$-values require caution: under an independently generated null, the usual clustered-normal rule rejects 24.1% of the time. Calibrating the absolute-$z$ threshold restores the primary null rejection rate to 4.65% (95% Monte Carlo CI $[3.81, 5.66]$%), but the observed post-law increase does not cross that threshold. At $B = 2{,}000$ per effect-size point, the calibrated 80%-power minimum detectable sustained decline is 80.0%; a Poisson sensitivity gives 64.1%. The paired Wilcoxon power calculation is not reported because the archived diagnostic drops zero-baseline events without a declared bootstrap rule. The functional conclusion is therefore

narrower than a claim of no effect: we observe no complaint reduction, but the design rules out only an extremely large sustained decline.

## Topic placebo: Telegram–Gov.News gaps

Figure 12 restates the topic placebo of Figure 7 as a single difference series per topic, so that all four topics sit on one shared axis and can be compared in sign and trend.

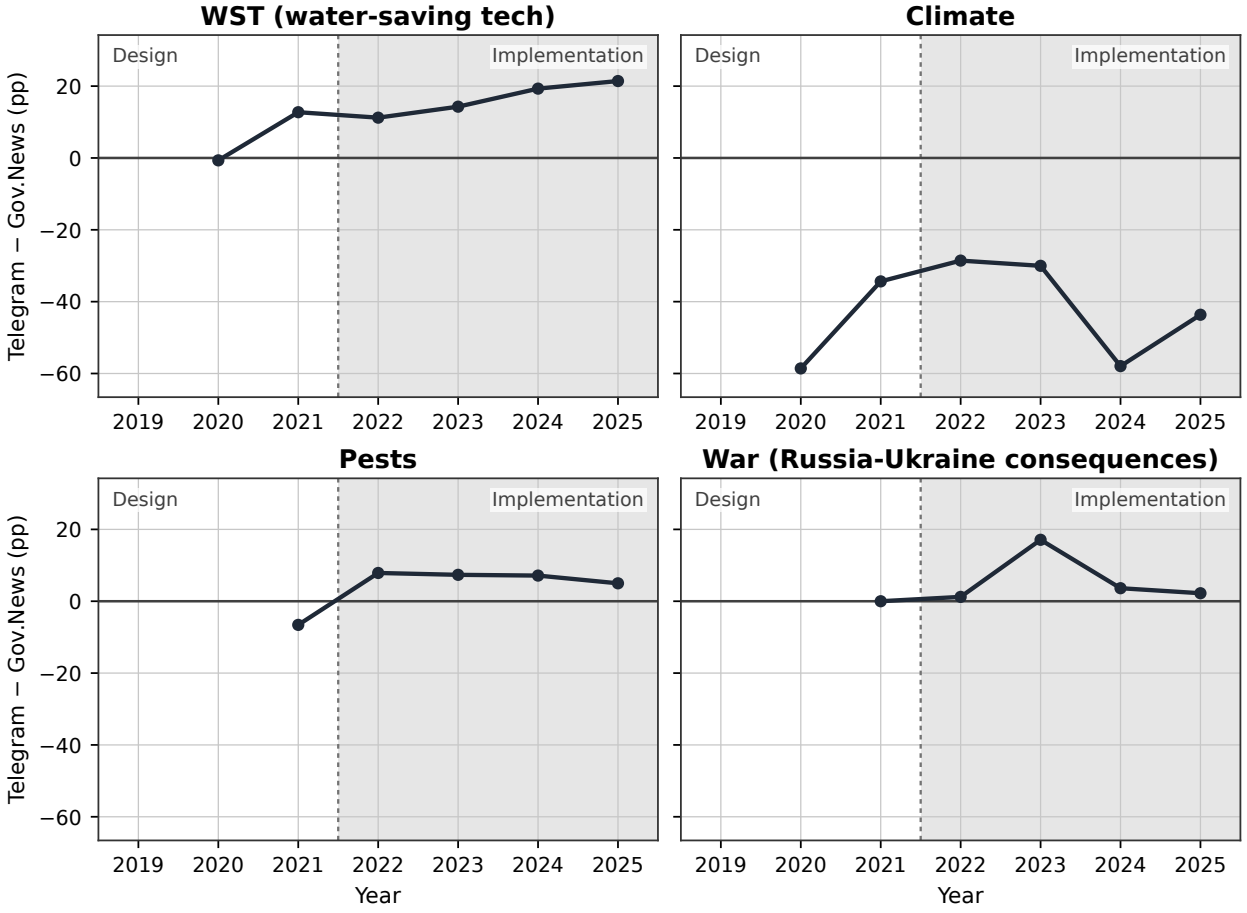



Figure 12 Telegram−Gov.News gap in topic-specific problem-mention rates, in percentage points on a single shared axis (positive = farmers report the topic's problems more often than the state outlet). Only WST shows a gap that opens in 2021 and widens through the implementation phase (from +11.2 pp in 2022 to +21.4 pp in 2025). Adverse weather/climate reverses the sign in every year (from −28.6 to −57.9 pp over 2021–25; the 2020 Telegram base is a single message), pests holds a roughly flat +5 to +8 pp through implementation, and the war topic shows a single transient spike in 2023. The dashed line marks the 2021.5 design/implementation split; the implementation window is shaded.

# B Clustering: Algorithm Sweep and Stability

**Construction.** Each problem is described by a 15-signal profile: design- and implementation-phase rates for laws, Gov.News, Telegram, donors, and Kun.uz, plus the corresponding phase shifts. For each source and phase, we first retain texts that mention at least one of the 14 canonical problems (distinct acts for laws). Attention to a given problem is the share of those texts that mention it. Because texts can mention several problems, these rates do not sum to one. The design/implementation denominators are 38/30 distinct law acts, 7/45 Gov.News articles, 25/1,035 Telegram messages, 43/57 donor documents, and 4/12 Kun.uz articles. The 25 design-phase Telegram messages are retained for the descriptive profiles and typology but excluded from the phase-level farmer-voice regression because the sample is too thin. These restricted denominators serve the typology and the agenda-persistence profiles (Table 5); the coverage rates reported elsewhere (Section 3, Figure 3) use each channel's full phase corpus (WST-positive texts, all donor documents, and for laws all problem links). Only the five phase-shift signals depend on the denominator choice; the ten within-phase level rankings do not. With 14 problems and small design-phase samples for Kun.uz and Gov.News, the typology is descriptive rather than a population estimator.

The five-profile policy-learning typology in Section 4.4 is obtained by ranking each of the 15 channel signals across the 14 WST problems, z-scoring the ranks, and clustering the resulting rank-profile matrix under cosine distance. Table 12 reports the robustness of this solution.

The cosine-average $k = 5$ solution has a silhouette (Rousseeuw, 1987) of 0.431, and $L_2$-normalized Ward and $L_2$-normalized $k$-means recover the same partition. Cosine-complete linkage at $k = 5$ produces a different one (adjusted Rand index (Hubert and Arabie, 1985) 0.662; silhouette 0.414), so agreement is substantial across three specifications but not universal (Panel A). In the sweep, the maximum silhouette at $k = 6$ is a three-way tie among cosine-average, cosine-complete, and cosine-single linkage (0.440), compared with 0.431 at $k = 5$ and 0.361 at $k = 4$; we retain $k = 5$ as the substantive summary, and the sweep does not single it out.

A leave-one-signal-out check (drop each of the 15 signals, re-rank, and re-cluster) recovers the partition exactly in 9 of 15 drops, with a mean adjusted Rand index of 0.868. In the feature-perturbation bootstrap (Panel B), mean ARI is 0.798; 45.2% of replicates recover the main structure (ARI $\geq 0.8$), and 44.0% recover the partition exactly. Within-profile co-membership is 0.999 for state learning, 0.993 for abandoned/dormant, 0.924 for governance failures, 0.918 for decay/coercion, and 0.696 for donor-mediated. With $n = 14$, these checks support a moderately stable descriptive typology rather than inferential coverage.

As a denominator sensitivity, replacing the restricted Kun.uz denominator with all WST-positive Kun.uz articles (248 in design; 184 in implementation) changes the $k = 5$ partition (silhouette 0.295; ARI 0.622 against the reported partition): equipment breakdown and faulty installation join the abandoned/dormant group. This is consistent with the small restricted Kun.uz samples and is why we read the boundaries involving these problems with caution.

**Channel-blind clustering.** Re-running the Figure 5 pipeline after dropping one channel at a time gives ARI 1.00 without laws, 0.79 without Gov.News, 0.66 without Telegram, 0.66 without donors, and 0.71 without Kun.uz. Dropping Telegram and Kun.uz together gives

Table 12 Stability of the $k$=5 cosine rank-profile typology ($n$=14 WST problems, 15 signals). Panel A compares four $k = 5$ clustering specifications on the same rank-profile matrix (adjusted Rand index, ARI, against the cosine-average solution; silhouette under cosine distance). Panel B reports a feature-perturbation bootstrap ($B$=1000 replicates, Gaussian noise of sd 0.25 on the z-scored profiles), including mean ARI, the share recovering the main structure (ARI $\geq 0.8$), exact recovery, and mean within-profile co-membership.

| *Panel A: multi-method agreement ($k$=5)* | | |
|---|---|---|
| Clustering specification | ARI vs. cosine-average | Silhouette |
| cosine-average linkage | 1.000 | 0.431 |
| cosine-complete linkage | 0.662 | 0.414 |
| Ward ($L_2$-normalized rows) | 1.000 | 0.431 |
| $k$-means ($L_2$-normalized rows) | 1.000 | 0.431 |
| *Panel B: feature-perturbation bootstrap ($B$=1000, sd 0.25)* | | |
| Statistic | Value | |
| Mean ARI vs. cosine-average | 0.798 | |
| Replicates with ARI $\geq 0.8$ | 45.2% | |
| Replicates with exact recovery (ARI = 1.0) | 44.0% | |
| *Mean within-profile co-membership* | | |
| donor-mediated loop ($n$=4) | 0.696 | |
| state-learning loop ($n$=2) | 0.999 | |
| governance failures ($n$=3) | 0.924 | |
| decay/coercion ($n$=2) | 0.918 | |
| abandoned/dormant ($n$=3) | 0.993 | |

ARI 0.40. The partition does not require the law signals alone, but it changes when any of the other single channels is removed and changes sharply when both public-facing channels are removed; the profile labels summarize joint trajectories across channels.

# C Expert Ratings: Elicitation and Triangulation

We rate each of the 14 problems on three criteria through a pairwise comparison game played by an expert rater, adapting the pairwise coding framework of Carlson and Montgomery (2017). The three criteria are observability (how legible the problem is to the central government), solvability (how tractable an effective response is), and donor-pressure sensitivity. The comparisons are fit with the Davidson tie-extended Bradley–Terry model (Davidson, 1970) with a Laplace posterior, yielding problem ranks (1 = hardest) with 80% credible intervals. The rater is one of the authors; the credible intervals therefore quantify uncertainty in the fitted one-rater model and say nothing about between-rater reliability. As triangulation, three archived headless LLM-rating sessions reproduce the human rankings with Spearman correlations of 0.71–0.94 for observability, 0.46–0.71 for solvability, and 0.62–0.73 for donor pressure; solvability is the weakest criterion. One session omitted the per-problem criterion briefs. The saved sessions preserve pairwise choices and fitted ranks, but the deleted runner's exact outer prompt wrapper and API settings are unavailable, so this triangulation is reproducible from judgments onward rather than from generation. In the human browser interface, pair order is randomized at presentation; the deleted LLM harness does not permit the same check. The rater had seen earlier channel-coverage results; the ratings, however, were elicited without reference to them, and the pairwise protocol (69–121 paired judgments per criterion, some pairs presented more than once, including ties and aggregated jointly by the Bradley–Terry fit) leaves little room to steer the resulting ranking toward an anticipated pattern.

# D Correlation-Difference Tests across Sources

To ask which sources align with implementation-phase law content, we form a public-media bundle (Telegram and Kun.uz) and a donor bundle, excluding Gov.News as a state-output channel, report rank-based partial correlations of Telegram and the donor bundle with law content, each conditional on the other, and decompose each correlation by problem. In the attention-trajectory figures, the attention-from-below measure takes the larger of the Telegram and Kun.uz within-source percentiles; the implementation-phase alignment between the two channels that motivates this composite is reported in Section 4.2.

The cross-channel analysis of Section 4.2 and the bundle diagnostic in Section 4.3 describe how strongly each source aligns with law content and with the other channels, judging each correlation against a zero null. A sharper question is whether two sources are *statistically distinguishable*: is one source's alignment reliably larger than another's ($\rho_1 > \rho_2$), rather than merely one clearing significance and the other not? Because the difference between a significant and a non-significant correlation need not itself be significant (Gelman and Stern, 2006), we test the difference directly. The same caution applies to the problem-level

Poisson panel, whose channel-swap test of the donor–voice coefficient difference is reported in Appendix E.

Each comparison involves two dependent, *overlapping* Spearman correlations that share one variable (the anchor) over the same 14 WST problems. We use the Williams (1959) test for two dependent overlapping correlations, computed on ranks, together with a joint percentile bootstrap ($B = 10{,}000$) that resamples the 14 problems and applies the same index set to every channel, recomputing $\Delta = \rho_1 - \rho_2$ on each replicate. The decision rule is fixed in advance: a pair is judged distinguishable only if Williams $p < 0.05$ *and* the bootstrap 95% CI for $\Delta$ excludes zero. We validated the Williams implementation by Monte-Carlo simulation. Under equal population correlations its empirical size is 0.047–0.053 against a nominal 0.05, with high power when the correlations differ, and it agrees throughout with the modified $z$ of Steiger (1980). Because that simulation uses large continuous samples, it validates the implementation rather than guaranteeing the asymptotic test at $n = 14$ with ties; the distribution-free bootstrap carries the small-sample inference, which is why we report a difference only when both criteria agree.

Two results follow (Table 15). First, the pre-specified law-alignment contrasts are all null: at $n = 14$ we cannot distinguish how well law content tracks the donor bundle from how well it tracks citizen voice (Telegram), in either policy phase or under a combined Telegram+Kun.uz public bundle; during implementation the donor and Telegram correlations are essentially identical ($\rho = +0.505$ and $+0.501$, $\Delta = +0.004$, Williams $p = 0.990$). The pairwise law-tracking gaps that the descriptive figures might suggest are not resolvable at this sample size. Second, the formal block's internal coherence *is* distinguishable from its link to citizen voice: the donor institutions agree with one another far more than any of them agrees with Telegram, and in the strongest cases (e.g. the implementation-phase UN–Research vs. UN–Telegram contrast) this difference survives both the bootstrap and a Bonferroni correction for the six comparisons. At the problem level, the wedge documented in the main text is therefore a separation of problem schemas (a coherent formal block and an external citizen-voice channel) rather than a graded ranking of which source the law answers; the ranking appears below the problem level, where the law's version of a problem follows the donor version (Section 4.3, Appendix E). Panel A is reproducible at `kunuz_recovery/src/run_h3_typology_kun_recovery.py` and Panel B at `appendix_checks/src/run_appendix_checks.py`.

Table 13 decomposes the two law correlations of Section 4.3. Donor and Telegram attention are uncorrelated across the 14 problems ($\rho = -0.02$), so conditioning on one source does not reduce the other's correlation with law content; only the donor interval excludes zero. The contributions of single problems (products of centered ranks, which sum to the rank numerator) show what the equal headline values conceal. For Telegram, high cost and unpaid subsidies contribute $+0.32$ of the $+0.50$, the seven locally revealed problems $+0.06$ in total, and forced adoption, the problem farmers raise most among those the law never touches, $-0.09$. For donors, the agreement is spread evenly across the Telegram halves and includes shared neglect: forced adoption contributes $+0.11$ because donors and the law both leave it aside, and weed infestation, second-lowest on both channels with zero laws, contributes $+0.13$ to both correlations. The law's agreement with donors therefore holds whether or not farmers raise a problem, whereas its agreement with Telegram is concentrated in the problems donors also flag. The computation is reproducible at `appendix_checks/src/decompose_law_alignment.py`.

Table 13 Where the two implementation-phase law correlations come from. Spearman correlations of problem-targeted law counts with Telegram and with the donor bundle, 2022–2025: across all 14 problems, conditional on the other source (problem-bootstrap 95% intervals, $B = 1{,}000$), within seven-problem subsets, and the contribution of the two problems every channel raises to each headline value (products of centered ranks, which sum to the rank numerator). The locally revealed group follows Hypothesis 3 (forced adoption, contractor fraud, faulty installation, corruption/fund misuse, equipment breakdown, water timing mismatch, and low yield/profit); its complement is high cost/loan burden, unpaid subsidies, no training/advice, power cost & cutoffs, water supply shortage, soil/climate mismatch, and weed infestation. Halves are median splits on the other source's attention. Subset correlations are descriptive; bootstrap intervals on every between-subset difference include zero.

| Problem set | Laws with Telegram | Laws with donors |
|---|---|---|
| All 14 problems | +0.501 | +0.505 |
| Contribution of high cost and unpaid subsidies | +0.32 | +0.17 |
| Given the other source (partial) | +0.589 [−0.113, +0.906] | +0.592 [+0.073, +0.857] |
| Locally revealed problems (7) | +0.180 | −0.144 |
| Their complement (7) | +0.811 | +0.396 |
| Donor-high / donor-low halves | +0.685 / +0.296 | – |
| Telegram-high / Telegram-low halves | – | +0.757 / +0.764 |

Table 14 Pairwise alignment of donor, farmer, and legal problem agendas during implementation, by problem set. Cells report Spearman correlations between the donor bundle, Telegram, and problem-targeted law counts, with source-unit bootstrap 95% intervals in brackets, 2022–25.

| Problem set | Donors–Telegram | Laws–Telegram | Laws–donors |
|---|---|---|---|
| All 14 problems | −0.02 [−0.16, +0.13] | +0.50 [+0.26, +0.59] | +0.50 [+0.28, +0.70] |
| Without high cost and unpaid subsidies (12) | −0.25 [−0.41, −0.07] | +0.21 [−0.11, +0.37] | +0.40 [+0.15, +0.64] |
| Locally revealed set plus weed infestation (8) | −0.24 [−0.48, +0.04] | +0.29 [−0.11, +0.49] | +0.02 [−0.34, +0.39] |
| Locally revealed problems (7) | −0.54 [−0.76, −0.25] | +0.18 [−0.32, +0.41] | −0.14 [−0.60, +0.30] |

Notes: Farmer voice is Telegram. The donor bundle averages within-group problem-attention percentile ranks for development banks, UN organizations, and research organizations. Each bootstrap replicate independently resamples donor documents within group, Telegram messages, and distinct legal acts, then reconstructs the three problem profiles ($B = 20{,}000$; seed 20260826). Intervals condition on the fixed problem schema and classifications. The locally revealed problems are forced adoption, contractor fraud, faulty installation, corruption, equipment breakdown, water timing mismatch, and low yield/profit. Adding weed infestation gives the eight-problem sensitivity row. Exact enumeration of all 7! problem-label permutations gives $p = 0.236$ for the donor–Telegram correlation, so the subset results are descriptive.

Table 15 Are two sources *statistically distinguishable* in how strongly they align with a common reference, or only individually different from zero? Each row tests the difference between two dependent, overlapping Spearman correlations that share one variable (the anchor), over the same $n = 14$ WST problems, using Williams's test on ranks and a joint percentile bootstrap ($B = 10{,}000$, resampling problems) of $\Delta = \rho_1 - \rho_2$. A pair is declared *distinguishable* only if Williams $p < 0.05$ *and* the bootstrap 95% CI excludes 0; such rows are marked * (every starred row here also has a CI excluding 0; one unstarred row, Design DevB·Res vs DevB·Tel, has a CI excluding 0 but Williams $p > 0.05$). The public bundle combines Telegram and Kun.uz. Panel A is pre-specified and confirmatory; Panel B is exploratory (six comparisons, uncorrected for multiplicity, though the two largest survive a Bonferroni correction). Design-phase Telegram correlations rest on the 25 problem-bearing messages of 2021 (Table 16, notes) and are reported for completeness. Correlations, $\Delta$, and intervals are displayed to two decimals, rounded from the unrounded statistics.

| Phase | $\rho_1$ vs $\rho_2$ (anchor) | $\rho_1$ | $\rho_2$ | $\Delta$ | Williams $p$ | Boot. 95% CI |
|---|---|---|---|---|---|---|
| *Panel A: Law alignment (anchor = laws).* | | | | | | |
| Impl. | donor bundle vs Telegram | +0.50 | +0.50 | +0.00 | 0.990 | [−0.72, +0.77] |
| Design | donor bundle vs Telegram | +0.46 | +0.36 | +0.10 | 0.718 | [−0.61, +0.80] |
| Impl. | donor bundle vs public bundle | +0.50 | +0.37 | +0.13 | 0.720 | [−0.63, +0.85] |
| Design | donor bundle vs public bundle | +0.46 | +0.33 | +0.13 | 0.624 | [−0.56, +0.79] |
| *Panel B: Donor coherence vs. donor–Telegram (anchor = first donor).* | | | | | | |
| Design | DevB·UN vs DevB·Tel | +0.73 | +0.39 | +0.34 | 0.164 | [−0.27, +0.92] |
| Design | DevB·Res vs DevB·Tel | +0.78 | +0.39 | +0.39 | 0.089 | [+0.03, +0.91] |
| Design | UN·Res vs UN·Tel | +0.85 | +0.41 | +0.44 | 0.037* | [+0.06, +0.96] |
| Impl. | DevB·UN vs DevB·Tel | +0.79 | +0.15 | +0.65 | 0.060 | [−0.15, +1.42] |
| Impl. | DevB·Res vs DevB·Tel | +0.92 | +0.15 | +0.77 | 0.004* | [+0.15, +1.44] |
| Impl. | UN·Res vs UN·Tel | +0.90 | −0.20 | +1.10 | 0.000* | [+0.49, +1.64] |

# E Direct Tests of Donor Substitution: Specifications

This appendix collects the specifications behind the sub-aspect analyses of Sections 4.2 and 4.3 and the panel, model-comparison, observer-chain, and event-study tests of Section 4.3. All estimates use distinct legal acts, end every channel's series in 2025, and treat years before a channel begins as missing rather than zero.

**Formal-agenda updating.** For each channel we keep the texts that mention at least one of the 14 problems (distinct acts for laws) and compute, for each problem, the share of those texts that mention it; a text can mention several problems. This gives one 14-number profile per phase, and we correlate the profile before the cutoff with the profile after it, using Spearman's $\rho$ and, for reference, Pearson's $r$. The primary split is 2018–2021 versus 2022–2025; the boundary check compares 2018–2022 with 2023–2025. Table 16 reports both statistics, both cutoffs and the number of texts behind each profile for all five channels. The main text focuses on donors and laws because coverage across the phase boundary is more limited in the remaining channels. These comparisons do not test equality of attention levels or prove that a channel received no new information.

Table 16 Problem-attention profiles before and during implementation, all five channels and both cutoffs. Each cell correlates a channel's 14-problem attention profile before the cutoff with its profile after (Pearson $r$ and Spearman $\rho$, two-sided asymptotic $p$-values in parentheses). The last column counts the texts that mention at least one problem, before/after: documents for donors, distinct acts for laws, articles for Gov.News and Kun.uz, messages for Telegram. Table 5 in the main text reports the donor and law rows with problem-bootstrap intervals.

| Channel | Pearson $r$ | Spearman $\rho$ | Texts before/after |
|---|---|---|---|
| *Implementation starts in 2022 (design 2018–21, implementation 2022–25)* | | | |
| Donors | +0.961 ($p < 0.001$) | +0.843 ($p < 0.001$) | 43 / 57 |
| Laws | +0.932 ($p < 0.001$) | +0.890 ($p < 0.001$) | 38 / 30 |
| Gov.News | +0.605 ($p = 0.022$) | +0.563 ($p = 0.036$) | 7 / 45 |
| Telegram | +0.305 ($p = 0.289$) | +0.401 ($p = 0.156$) | 25 / 1,035 |
| Kun.uz | +0.121 ($p = 0.680$) | +0.083 ($p = 0.778$) | 4 / 12 |
| *Implementation starts in 2023 (design 2018–22, implementation 2023–25)* | | | |
| Donors | +0.961 ($p < 0.001$) | +0.857 ($p < 0.001$) | 65 / 35 |
| Laws | +0.880 ($p < 0.001$) | +0.716 ($p = 0.004$) | 48 / 20 |
| Gov.News | +0.703 ($p = 0.005$) | +0.777 ($p = 0.001$) | 27 / 25 |
| Telegram | +0.507 ($p = 0.064$) | +0.645 ($p = 0.013$) | 220 / 840 |
| Kun.uz | +0.292 ($p = 0.311$) | +0.426 ($p = 0.129$) | 10 / 6 |

Notes: The donor and law results hold under either boundary. Telegram shows a cross-phase association only when 2022 is included in the earlier period; with the 2022 cutoff its design profile rests on 25 messages. Kun.uz shows no cross-phase association under either cutoff, on 4 and 10 design articles. $p$-values are computed over $n = 14$ problems and make no allowance for the small design bases of the last three channels.

**Pairwise alignment by problem set.** Table 14 correlates the implementation-phase donor-bundle, Telegram, and law profiles across all 14 problems and three nested restrictions. The donor bundle is reconstructed as the mean of the development-bank, UN, and research-organization percentile-rank profiles in every bootstrap draw. Intervals independently resample donor documents within group, Telegram messages, and distinct legal acts ($B = 20{,}000$); the seven-problem label permutation instead permutes the problem labels of one vector. These procedures condition on the fixed problem schema and classifications.

**Problem×year panel.** Outcome: distinct-act problem-targeted law counts for problem $p$ in year $t$. Regressors are one-year-lagged within-year problem shares of donor documents and Telegram messages, z-scored, with year fixed effects and SEs clustered by problem. Because the chat begins in late 2020, 2020 has no lagged voice, leaving 70 observations over 2021–2025. The additive estimates are donor +0.407 (cluster $p = 0.015$, within-year permutation $p = 0.012$) and voice +0.197 (cluster $p = 0.216$, permutation $p = 0.194$). The exact $2^{14}$ channel-swap test of their difference gives $p = 0.409$ (Wald $p = 0.287$). Dropping the 14 structurally zero lagged-voice rows for 2021 leaves donor +0.406 ($p = 0.013$) and voice +0.198 ($p = 0.216$). In the interaction model, donor is +0.448 ($p = 0.029$), voice +0.156 ($p = 0.360$), and donor×voice is null ($p = 0.438$). Timing variants sharpen the picture: at lag two, voice is +0.475 (cluster $p = 0.070$) and donor +0.229 ($p = 0.133$); a specification whose regressor is the mean of the lag-one and lag-two attention shares (56 observations) gives donor +0.322 ($p = 0.050$) and voice +0.310 ($p = 0.119$); over 2023–2025, donor is +0.520 and voice +0.395 (cluster $p < 0.001$ for both; within-year permutation $p = 0.027$ and 0.083).

**Phase-level model comparison and transmission.** The 25 design-phase Telegram messages that mention at least one of the 14 problems remain in the descriptive profiles and typology, but their sample is too thin for a design-phase farmer-voice regression. Implementation counts are overdispersed, so plain Poisson likelihood-ratio tests overstate precision. Under negative-binomial likelihood-ratio, quasi-Poisson, and within-phase permutation inference, adding voice to donor-only gives $p = 0.010$, 0.008, and 0.006; adding donor to voice-only gives $p = 0.007$, 0.004, and 0.007. Negative-binomial AIC is 76.9 for the combined model, 81.5 for donor-only, 82.2 for voice-only, and 83.4 for the null. Donor-only and voice-only fits remain indistinguishable by the non-nested Vuong comparison. Transmission is OLS of within-year donor problem-mention share on lagged Telegram share, both standardized, with problem and year fixed effects and SEs clustered by problem: $b = +0.164$, 95% CI $[-0.09, +0.42]$, cluster $p = 0.203$. Within-year permutation ($p = 0.048$) and wild-cluster bootstrap ($p = 0.250$) disagree, so the relay question is unresolved.

**Observer chain and weekly event study.** Chain (56 observations, 2021–2024): Poisson with $t+1$ outcomes and log-exposure offsets, lagged complaint-rate control, year fixed effects, and SEs clustered by problem. Donor→law is $b = +0.326$ ($p = 0.120$; this stage is the panel's donor-only specification restricted to 2021–2024, hence the difference from +0.407). Using the full canonical Qwen observer series, law→next-year negative reports is $b = +0.200$ ($p = 0.169$), and the reduced donor→negative association is $b = -0.041$ ($p = 0.735$). A cleaned

direct-message-only Qwen sensitivity gives $b = +0.214$ (cluster $p = 0.118$, permutation $p = 0.0634$) and a reduced association of $b = -0.049$ ($p = 0.670$). The chain therefore does not establish mediation. The weekly study uses 63 implementation-phase act×problem events from 23 acts with complete ±26-week windows; a stacked Poisson includes event and calendar-quarter fixed effects and log weekly chat-volume offsets. Weeks are indexed from the enactment week: the reference base is weeks −26 to −9, the pre-window −8 to −1, the announcement window 0 to 4, and the post-window 5 to 26. Direct decree references, their two-hop reply threads (replies posted from one day before to 30 days after the referencing post), and twelve high-volume news accounts (each with at least 15 law-referencing posts, together 462 of 1,344) are removed from both numerator and exposure in the corresponding variants.

**Gauge validation.** The primary simulation draws once for each unique problem–calendar-week from a fitted NB2 null and maps that draw to every overlapping event-stack row. An independent $B = 2{,}000$ null sample calibrates the 95th percentile of absolute clustered $z$ at 3.668; the observed post coefficient has $z = 2.60$. Power is computed with $B = 2{,}000$ draws at each target rate ratio under the NB2 null and $B = 500$ under a Poisson null; two of 47,500 power-curve fits (19 effect-size points under each null) fail to converge (0.004%) and are retained as non-detections, and all 2,500 critical-value calibration fits and both independent null samples converge. The resulting rejection rates and minimum detectable declines, the positive-control clauses, and the importance-split interactions are reported in Appendix A.

**Sub-aspect classification.** Each of the fourteen problems is decomposed into five to eight sub-aspects, 94 codes in all, under one protocol: an independent discovery report per problem written from raw text and top-level tags only (GPT-5.6 Luna), adjudicated and frozen by GPT-5.6 Sol before any classifier output was inspected. The classified sample contains 2,235 source×problem units and 8,337 full-text chunks: Telegram message–problem units from 2022–2025 (capped at 200 messages per problem with seed 42, a cap that binds for high cost and subsidy nonpayment only), donor document– and law record–problem units through 2025 (not sampled, so the comparison is not fully phase matched), and Gov.News, Kun.uz, and FGD transcript units from 2022–2025. Qwen3.8-27B/NInfer with medium reasoning and GPT-5.6 Luna with high reasoning classify the full documents in 40,000-character chunks with 8,000-character overlap and no document-level character cap; the primary label is the liberal code union, not an adjudicated vote. Luna covers every chunk and supplies the label alone for the 159 chunks on which Qwen never returned a validated result, so the Qwen-primary check is not independent on those chunks; one donor chunk of garbled custom-font OCR on which neither coder could quote literally is recorded as empty. Empty classifications are retained as zero-assignment units, and same-act bilingual law records are collapsed by code union. Across the 2,181 units carrying both coders' labels, exact code-set agreement is 54.5% and mean Jaccard similarity 0.679 (45.0% and 0.611 excluding units empty under both coders). Source distributions normalize all assigned codes, and $B = 4{,}000$ source-unit bootstrap draws estimate base-2 Jensen–Shannon divergence: high cost, lack of training, and water-source unavailability were designated confirmatory and use 95% percentile intervals;

electricity supply, soil/climate suitability, and subsidy nonpayment use Bonferroni-adjusted familywise 95% intervals (soil/climate is descriptive at five law acts, and subsidy nonpayment has 3,959 valid draws because 41 resamples contain no donor assignments); problems with no law act carrying a sub-aspect are reported descriptively. The pooled test is a Poisson regression of the number of distinct law acts carrying sub-aspect $s$ of problem $p$ on the share of problem $p$'s donor assignments and Telegram assignments going to $s$, with problem fixed effects; $p$-values permute one regressor across sub-aspects within each problem (5,000 draws), the donor minus Telegram difference is tested by swapping the two shares sub-aspect by sub-aspect, and problems with no law act carrying any sub-aspect drop out. Letting both slopes differ between the six generic problems that reach law and the locally revealed problems, the donor slope falls from 6.27 to 1.98 on the locally revealed problems (within-problem permutation $p = 0.095$) while the Telegram slope does not rise detectably; the sign pattern holds under the Qwen-primary and Luna-only labels.

**Sub-aspect alignment.** Table 3 treats each sub-aspect's share of a source's assignments within its problem as that source's version of the problem. For a pair of sources, shares are demeaned within each problem and pooled across problems, and the statistic is the Pearson correlation of the pooled deviations; problems with fewer than three codes or with no assignment from either source are excluded, which is why the number of problems differs by pair. Inference permutes the second source's shares across sub-aspects within each problem (5,000 draws). Differences between two correlations that share a source are tested by swapping the two comparison sources' share vectors within problems. The unweighted mean of within-problem Spearman correlations gives the same ordering.

# F Illustrative Farmer Quotations

These quotations illustrate the WST problems the classifier identifies in the view-from-below sources; they validate the coding and are not study findings. From the FGDs/IDIs:

1. **High cost/loan burden (installation and loans)**
   (a) High Investment Burden: *"25 million per hectare... farmer paid 17 million with 20% interest. The interest is hurting us."*
   (b) Infrastructure Expenses: *"Had to install transformer for 25 million because drip irrigation requires 3-phase power"*
2. **Equipment breakdown**
   (a) Substandard Hoses: *"Their irrigation hoses are very poor quality, breaking down in less than a year."*
   (b) Inconsistent Quality Standards: *"If you use a high-quality drip system, there won't be problems, but the cheaper ones cause issues with leaks and blockages."*

From the Telegram chat:

1. **Contractor fraud**
   (a) Contractor Fraud: *"Chinese contractor promised drip irrigation... deceived for 500 million."*
2. **No training/advice**
   (a) No Technical Support: *"No specialists available, complete lack of knowledge"*
   (b) Technical Difficulties: *"Got close to 50 centners but couldn't work according to stated technology"*
   (c) Basic Knowledge Issues: *"Can weed killer be added to drip irrigation?"*

# G Problem Schema: 19-to-14 Crosswalk

**Validation detail.** The 100-text gold sample was drawn balanced between screen-positive and screen-negative texts rather than stratified by source; its composition is 87 Telegram messages, 9 Gov.News articles, 3 Kun.uz articles, and one FGD transcript, with the law and donor corpora not covered. The benchmark validates the final any-problem decision, not the preliminary WST screen. Differential error across sources could in principle contribute to apparent cross-source differences, a bias analogous to the one Ashwin et al. (2025) document for LLM coding of interview text. When we decompose error by source on the gold sample, error rates do not differ detectably (Fisher exact $p \geq 0.71$), but the sample's Telegram dominance leaves that test with little power for the official channels. The ensemble's problem-share ranking on the 87 Telegram texts correlates with the coder's at Spearman $\rho = 0.90$ across the twelve crosswalked buckets and 0.88 across the 19 precursor categories (`appendix_checks/src/benchmark_share_ranking.py`). A further caveat is that the benchmark was coded by a single expert, so there is no inter-rater statistic. A full gold-anchored re-estimation would require probability-sampled human labels on the law and donor corpora, which we leave to an extension. At the sub-aspect level, the two coders' exact code-set agreement and Jaccard similarity are reported in Appendix E.

The classification benchmark referenced in Section 3.2 was run on a precursor 19-category problem scheme; the 14 canonical problems used throughout the paper are a regrouping of that scheme, refined during iterative schema development. Table 17 gives the mapping. Two refinements bear on the argument: the precursor `corruption` category was split into *corruption/fund misuse* and *contractor fraud*, and `bureaucratic resistance` was narrowed to *forced adoption* (coercion). Because these two categories are refinements rather than direct benchmark categories, their problem-level accuracy is not separately validated.

Table 17 Crosswalk from the 19-category benchmark scheme to the 14 canonical problems used in the analysis.

| **19-category benchmark ID(s)** | **14 canonical problem** |
|---|---|
| `installation_cost` + `loan_problems` | High cost/loan burden |
| `subsidy_issues` | Unpaid subsidies |
| `equipment_quality` + `system_performance` + `parts_unavailable` | Equipment breakdown |
| `installation_problems` | Faulty installation |
| `power_supply` | Power cost & cutoffs |
| `water_supply` | Water supply shortage + water timing mismatch |
| `corruption` | Corruption/fund misuse + contractor fraud |
| `bureaucratic_resistance` | Forced adoption (coercion) |
| `knowledge_how_to_use` + `knowledge_value` + `poor_service` | No training/advice |
| `crop_compatibility` | Soil/climate mismatch |
| `weed_problems` | Weed infestation |
| `not_sufficient_yield_profit` | Low yield/profit |
| `basic_facilities`, `field_operations` | — (excluded; near-zero signal) |

# H Classification Prompts

The full prompt texts are provided in a separate supplementary file, *Supplementary Material: Classification Prompts*. The supplement reproduces verbatim the recoverable production prompts behind the reported results: the binary WST screens and the 14-problem ensemble prompts for the farmer, media, law, and donor corpora (Section 3.2), the marginal-loss verification templates, and the blame, sub-aspect, sentiment, and topic-placebo prompts. For the rating game, it prints the recoverable task content and explicitly records that the headless LLM runner's exact outer wrapper is unavailable; the saved judgments remain reproducible downstream. The 19-category benchmark scheme's crosswalk to the 14 canonical problems is Table 17.